Time Series Modeling

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Introduction

Project Overview

The goal of this project is to analyze and model the time series dynamics of *Life Expectancy* and *Unemployment* using various statistical and econometric techniques. These response variables exhibit interdependent fluctuations over time, making them suitable for vector autoregressive (VAR) modeling. The objective is to identify the best predictive models using autoregressive (AR), autoregressive distributed lag (ARDL), and VAR techniques, assessing their performance using various diagnostic tools.

Response and Predictor Variables

In this analysis, we consider the following variables:

Response Variables:

1. **Life Expectancy** (*y1t*):
   - Represents the average number of years a newborn is expected to live, given current mortality rates.
   - Higher values indicate better public health conditions and quality of life.
2. **Unemployment Rate** (*y2t*):
   - Measures the percentage of the labor force that is actively seeking but unable to find employment.
   - Higher values signal economic distress, while lower values indicate a strong job market.

Predictor Variables:

1. **Health Expenditure (%)**:
   - Represents the percentage of a country's GDP spent on healthcare.
   - Affects both life expectancy (positively) and unemployment (potentially due to healthcare-related job creation).
2. **Prevalence of Undernourishment (%)**:
   - Measures the percentage of the population with insufficient caloric intake.
   - Higher values are expected to negatively impact life expectancy and potentially influence unemployment due to workforce productivity loss.
3. **Education Expenditure (%)**:
   - Represents the percentage of GDP allocated to education.
   - Can positively affect life expectancy by improving public health knowledge and also impact unemployment by increasing job opportunities.

Methodology

This project follows a structured approach to model selection and evaluation:

1. **Descriptive Analysis**:
   - Visual and statistical exploration of the data, including histograms, boxplots, scatterplots, correlation plots, and summary statistics.
2. **Time Series Diagnostics**:
   - Examining stationarity using ACF/PACF plots.
   - Transforming data if necessary (e.g., differencing).
3. **AR Models**:
   - Fitting two AR(p) models to each response variable and assessing their residual diagnostics.
   - Comparing models using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).
4. **ARDL Models**:
   - Identifying significant predictors and fitting ARDL(p, q) models.
   - Evaluating forecasting accuracy and comparing performance metrics.
5. **VAR Model**:
   - Modeling the interaction between *Life Expectancy* and *Unemployment*.
   - Assessing Granger causality, impulse response functions (IRF), forecast error variance decomposition (FEVD), and stability through CUSUM plots.
6. **Model Evaluation and Forecasting**:
   - Training-testing performance assessment.
   - Computing multi-step-ahead forecasts.
   - Choosing the best predictive model for each response variable.

Expected Outcomes

By applying these econometric techniques, we aim to: - Understand the time-dependent relationships between health, education, nutrition, and economic conditions. - Identify the best-performing forecasting model for *Life Expectancy* and *Unemployment*. - Determine whether a VAR model captures interdependencies more effectively than univariate approaches.

The findings of this study will contribute to better policymaking and economic planning, helping decision-makers understand the interplay between public health spending, nutrition, education, and labor market dynamics.

```r
df<- read.csv("life.csv", header= TRUE, sep = ",")
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(forecast)
library(tseries)
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```r
library(dynlm)
```

```
## Loading required package: zoo
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(ggplot2)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(vars)
```

```
## Loading required package: MASS

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select

## Loading required package: strucchange

## Loading required package: sandwich

## Loading required package: urca

## Loading required package: lmtest
```

```r
library(urca)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
library(strucchange)
```

```r
# Load data
df <- read.csv("life.csv")

# Corrected variable names based on actual dataset
selected_vars <- c("Life.Expectancy.World.Bank", "Prevelance.of.Undernourishment",
                   "Health.Expenditure..", "Unemployment", "Education.Expenditure..")

# Ensure the selected columns exist in the dataset
selected_vars <- selected_vars[selected_vars %in% colnames(df)]
```

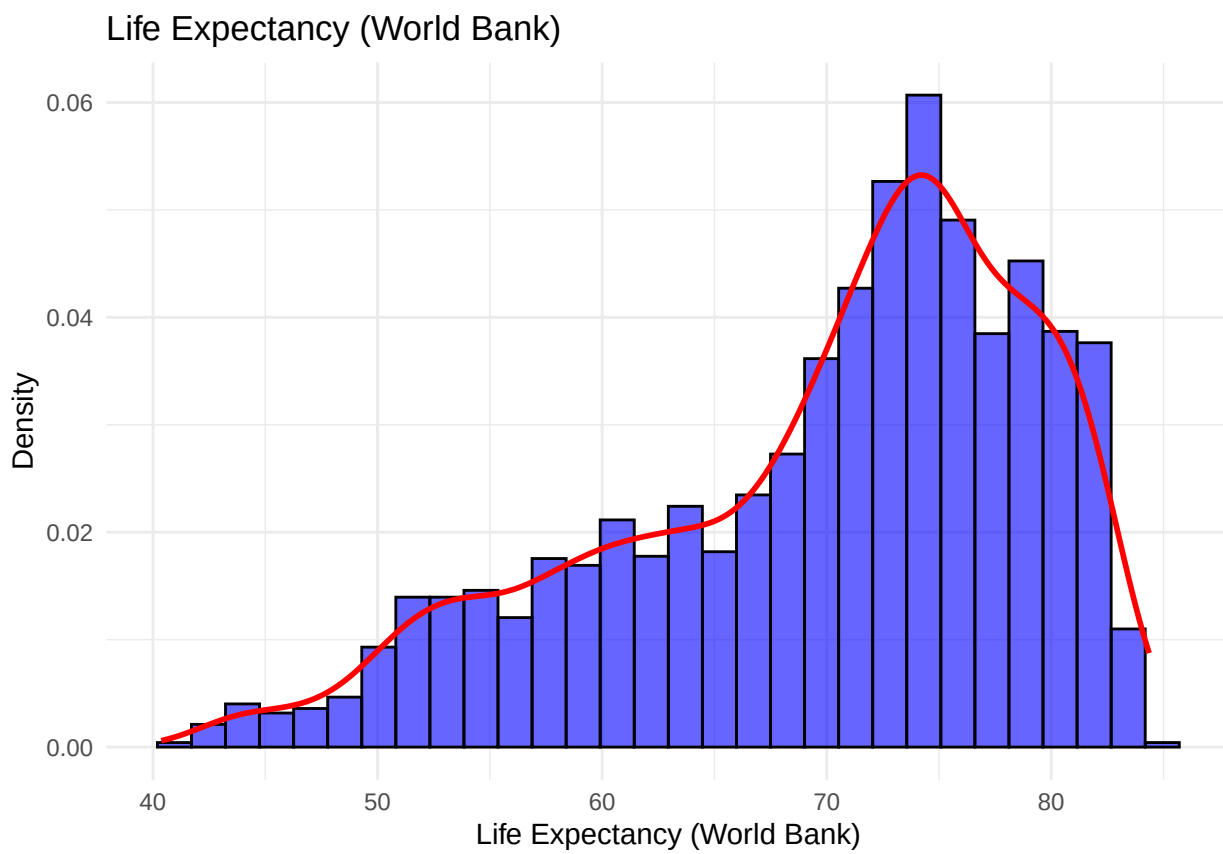
```r
# Function to compute five-number summary
five_num_summary <- function(var) {
  return(fivenum(var, na.rm = TRUE))  # Remove NA values
}

# Apply function to selected variables
summary_results <- lapply(df[selected_vars], five_num_summary)

# Convert to a data frame for better readability
summary_df <- data.frame(
  Variable = selected_vars,
  Min = sapply(summary_results, function(x) x[1]),
  Q1 = sapply(summary_results, function(x) x[2]),
  Median = sapply(summary_results, function(x) x[3]),
  Q3 = sapply(summary_results, function(x) x[4]),
  Max = sapply(summary_results, function(x) x[5])
)

# Print results
print(summary_df)
```

```
##                                                    Variable       Min
## Life.Expectancy.World.Bank         Life.Expectancy.World.Bank 40.369000
## Prevelance.of.Undernourishment Prevelance.of.Undernourishment  2.500000
## Health.Expenditure..                     Health.Expenditure..  1.263576
## Unemployment                                     Unemployment  0.100000
## Education.Expenditure..               Education.Expenditure..  0.850320
##                                     Q1    Median        Q3      Max
## Life.Expectancy.World.Bank     63.640000 72.168500 76.812000 84.35634
## Prevelance.of.Undernourishment  2.500000  6.200000 14.800000 70.90000
## Health.Expenditure..            4.204817  5.892352  8.119234 24.23068
## Unemployment                    3.731000  5.920000 10.100000 37.25000
## Education.Expenditure..         3.135635  4.371465  5.519840 23.27000
```

```r
# Load data
df <- read.csv("life.csv")

# Select relevant variables
selected_vars <- c("Life.Expectancy.World.Bank", "Prevelance.of.Undernourishment",
                   "Health.Expenditure..", "Unemployment", "Education.Expenditure..")

# Ensure columns exist in the dataset
df_selected <- df[selected_vars]

# HISTOGRAMS WITH FITTED DENSITY CURVE
hist_density_plot <- function(var, title) {
  ggplot(df, aes(x = var)) +
    geom_histogram(aes(y = ..density..), bins = 30, fill = "blue", alpha = 0.6, color = "black") +
    geom_density(color = "red", size = 1) +
    labs(title = title, x = title, y = "Density") +
    theme_minimal()
}

# Generate individual histograms
```

```r
hist_density_plot(df$Life.Expectancy.World.Bank, "Life Expectancy (World Bank)")
```

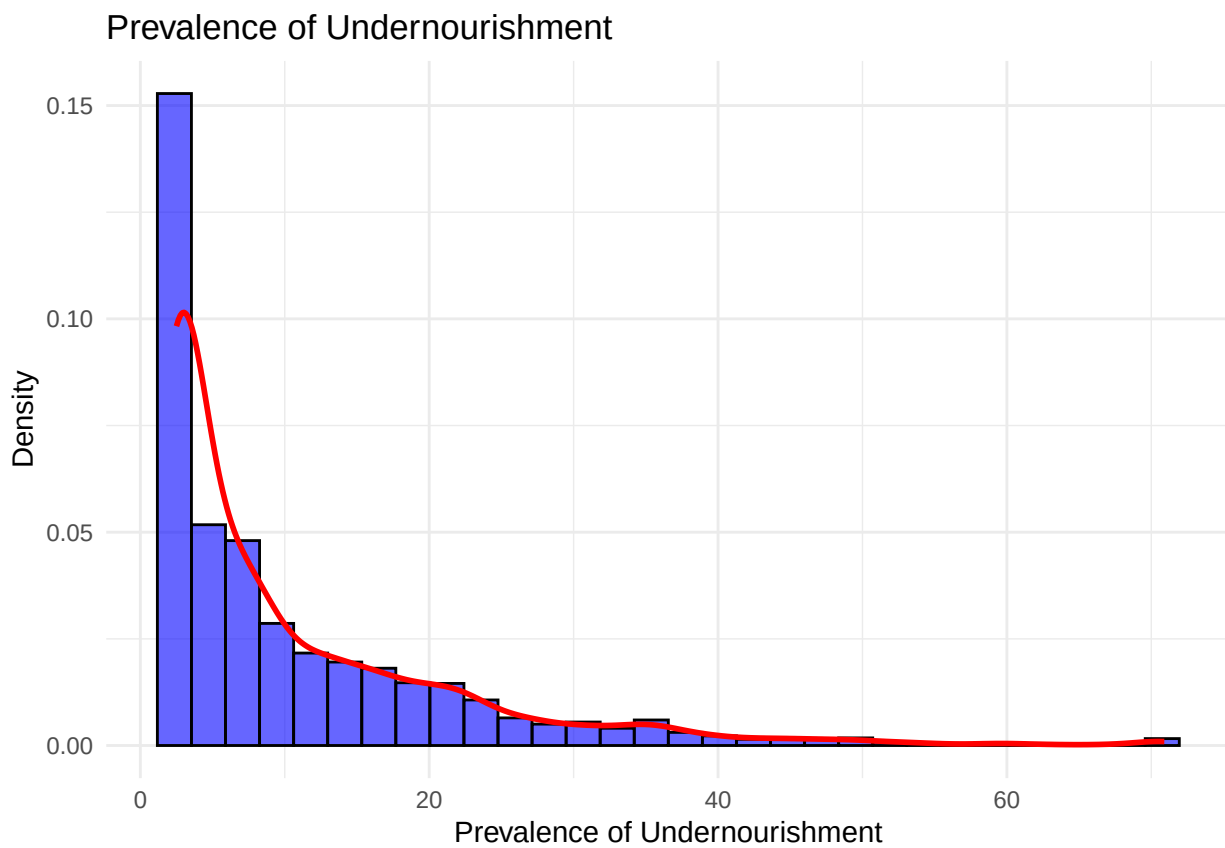
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Removed 188 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 188 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```r
# This histogram shows the distribution of life expectancy across countries. The red density curve
# overlays a smoothed estimate of the distribution, helping to visualize the shape of the data.

hist_density_plot(df$Prevelance.of.Undernourishment, "Prevalence of Undernourishment")
```

```
## Warning: Removed 684 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 684 rows containing non-finite outside the scale range
```

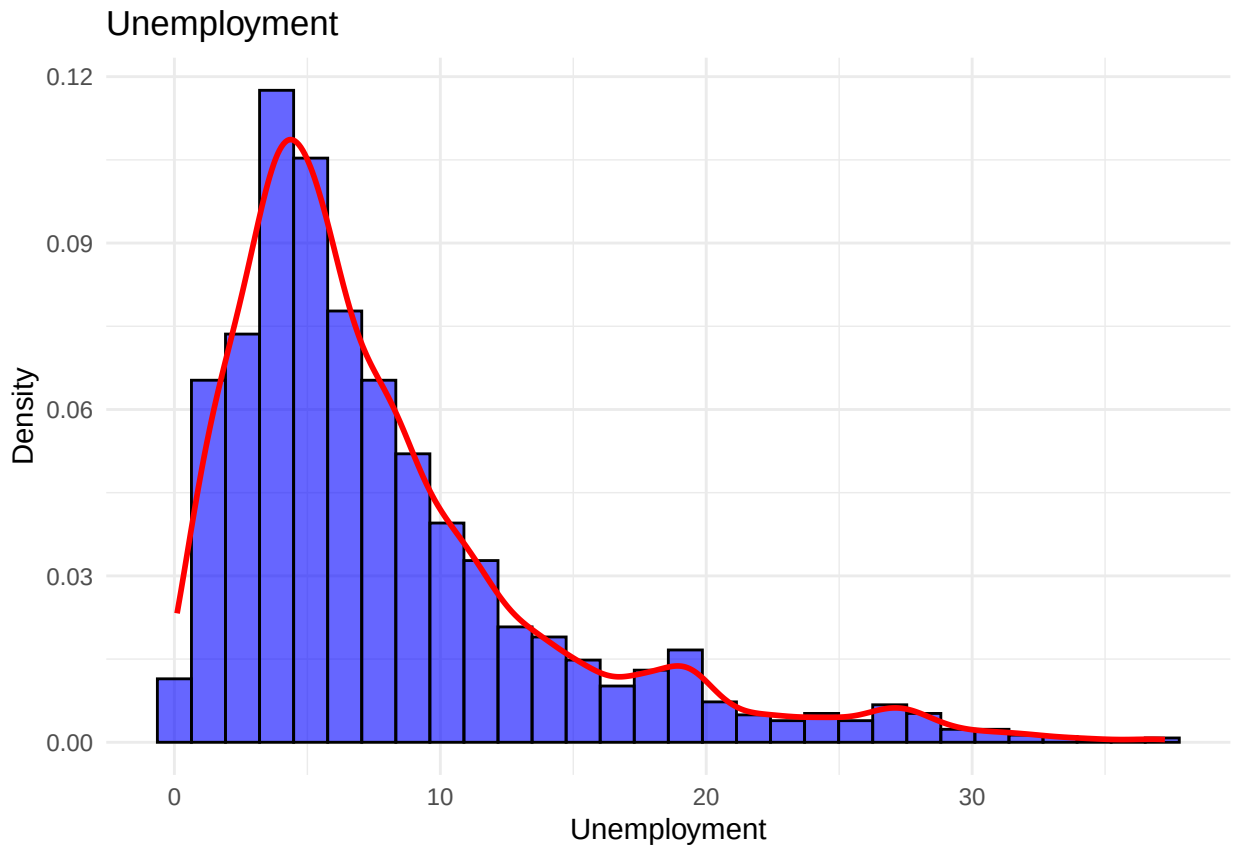
```
## (`stat_density()`).
```




```r
# This histogram displays the percentage of the population suffering from undernourishment.
# The red density curve helps illustrate whether the data is skewed or normally distributed.

hist_density_plot(df$Unemployment, "Unemployment")
```

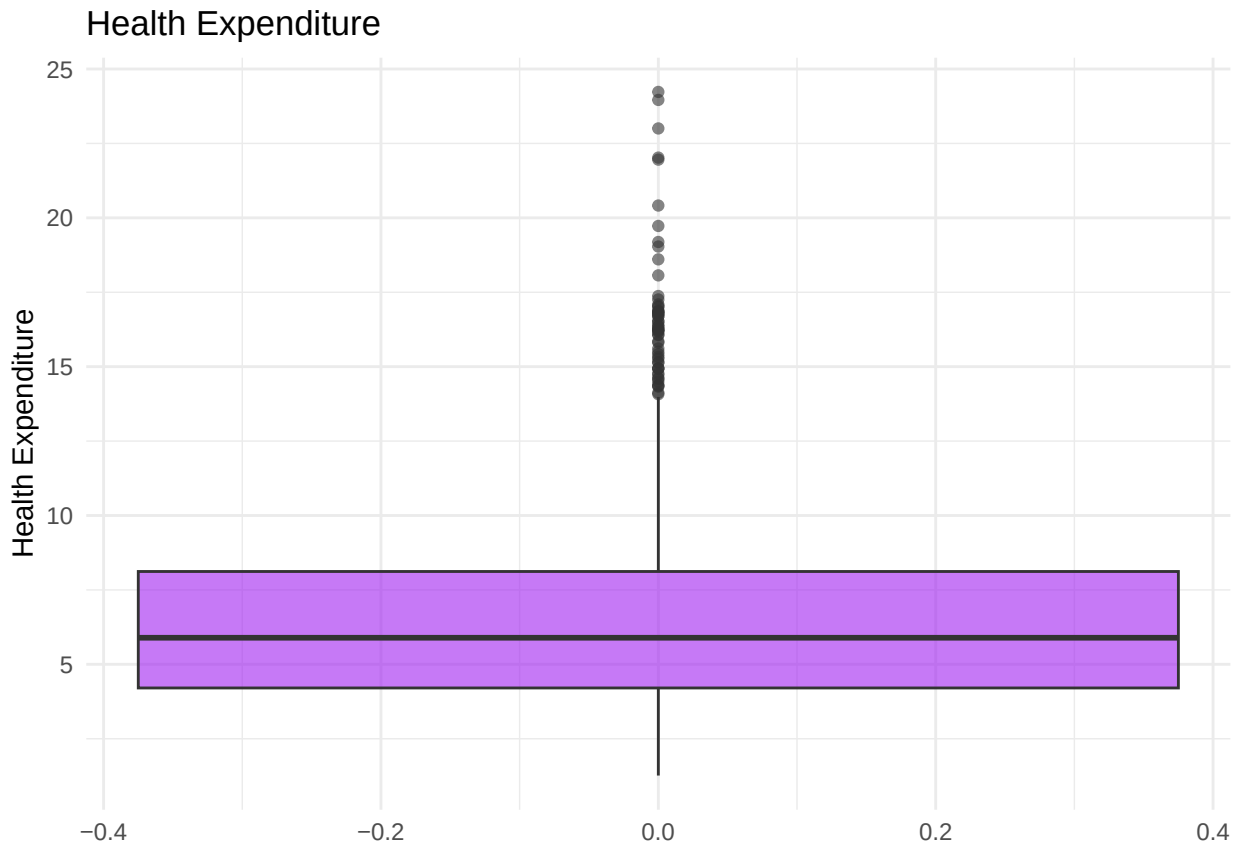
```
## Warning: Removed 304 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

```
## Warning: Removed 304 rows containing non-finite outside the scale range
## (`stat_density()`).
```



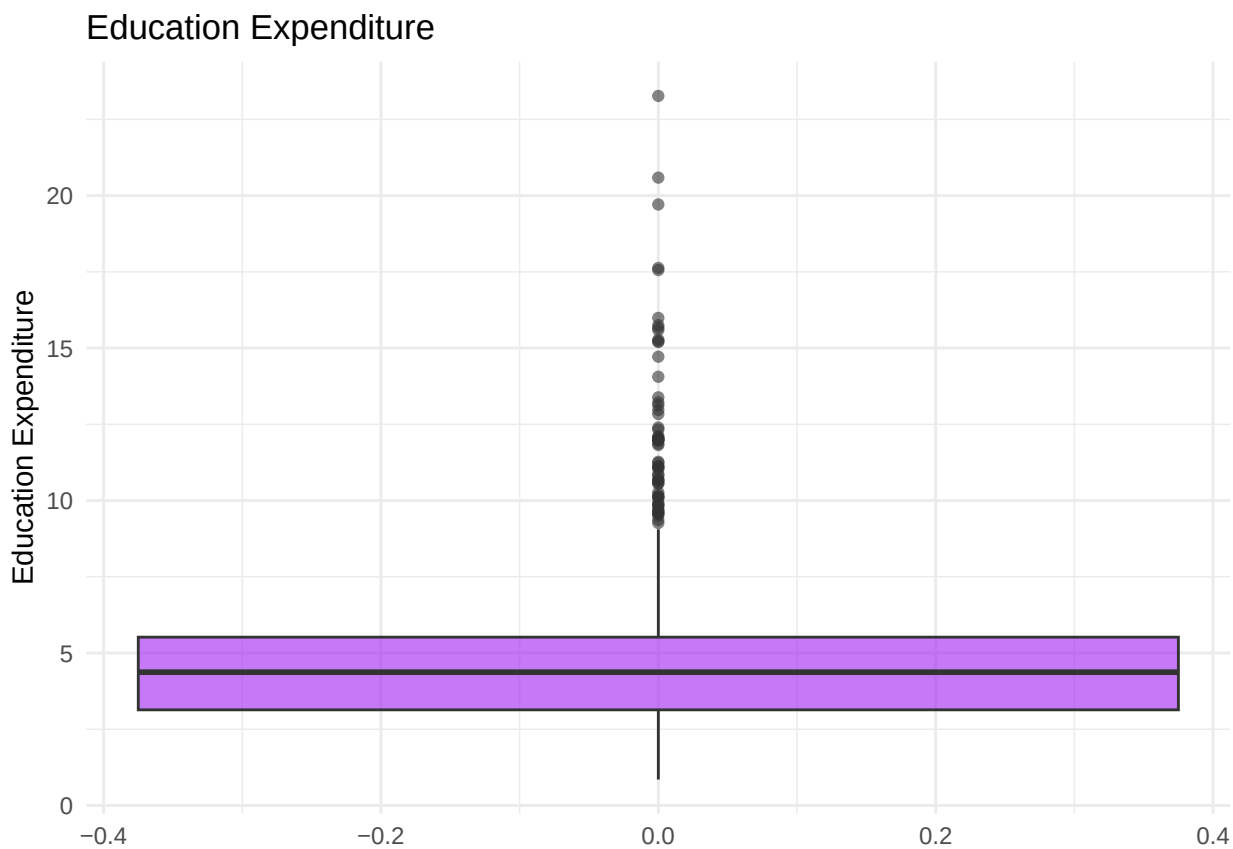

```
# This histogram represents the distribution of unemployment rates across countries.
# The density curve helps identify the most frequent unemployment levels and any outliers.

# BOX PLOTS
boxplot_plot <- function(var, title) {
  ggplot(df, aes(y = var)) +
    geom_boxplot(fill = "purple", alpha = 0.6) +
    labs(title = title, y = title) +
    theme_minimal()
}

# Generate individual boxplots
boxplot_plot(df$Health.Expenditure.., "Health Expenditure")
```

```
## Warning: Removed 180 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

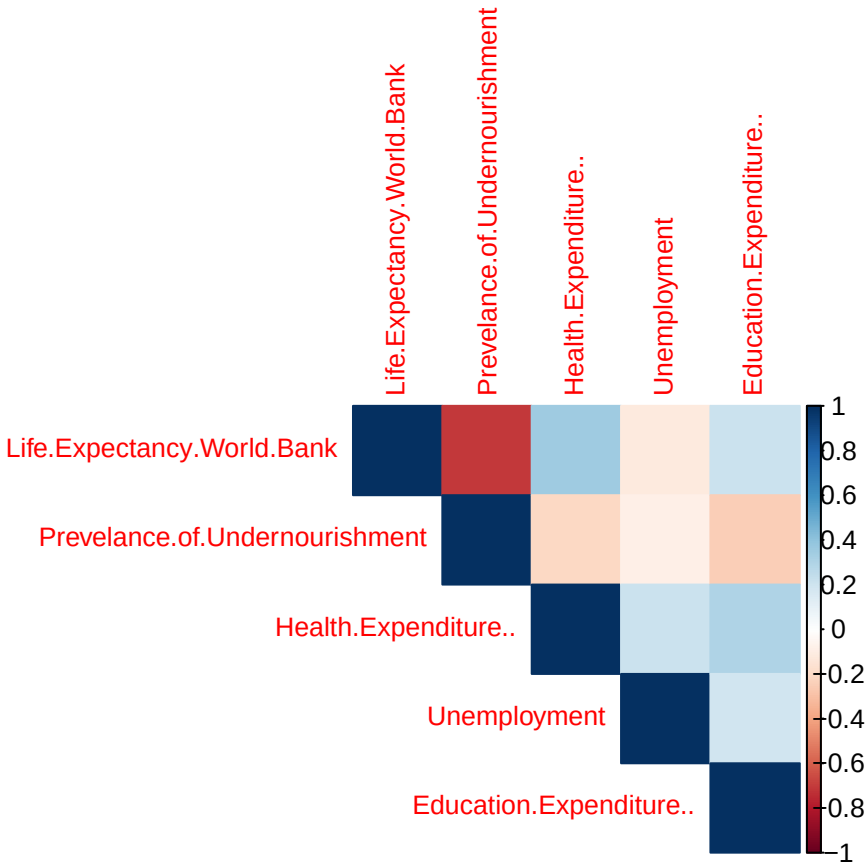



```r
# This boxplot visualizes the spread of health expenditure data, highlighting the median,
# interquartile range, and potential outliers.

boxplot_plot(df$Education.Expenditure.., "Education Expenditure")
```

```
## Warning: Removed 1090 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

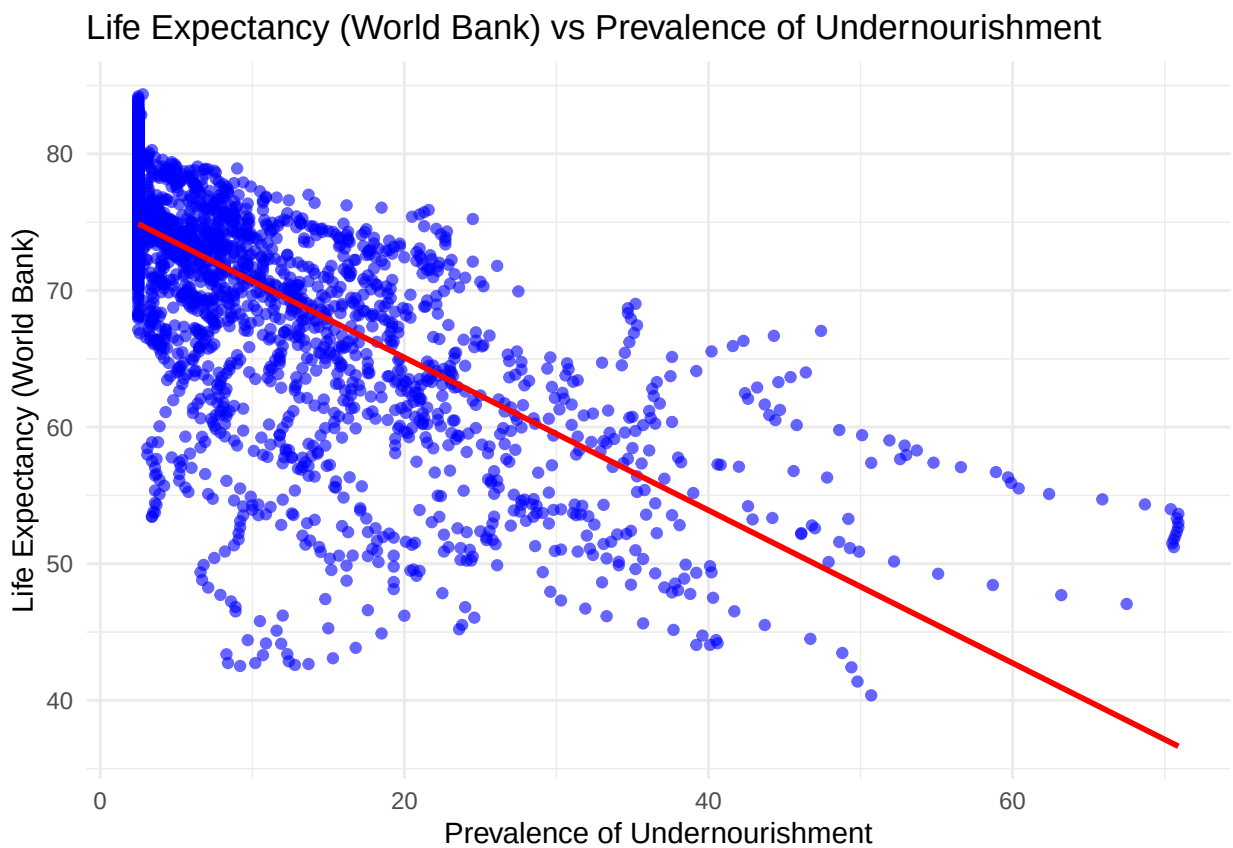



```
# This boxplot shows the variation in education expenditure across countries, allowing
# us to detect high-spending and low-spending regions.

# CORRELATION PLOT (Generated Individually)
corr_matrix <- cor(df_selected, use = "complete.obs")
corrplot(corr_matrix, method = "color", type = "upper", tl.cex = 0.8)
```



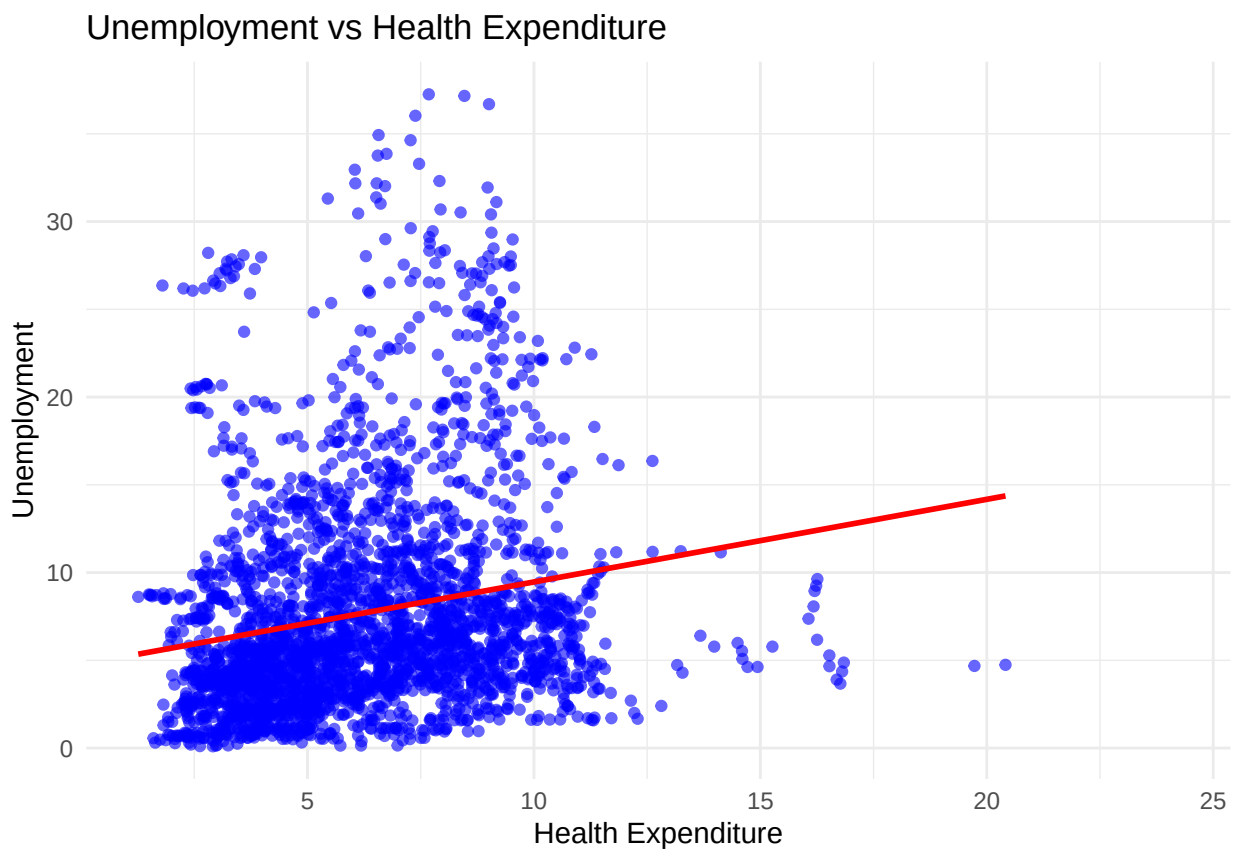

```r
# The correlation plot illustrates relationships between all selected variables. Darker colors indicate
# stronger correlations, with red representing negative correlations and blue representing positive cor

# SCATTERPLOTS WITH REGRESSION LINE
scatterplot_plot <- function(xvar, yvar, xlabel, ylabel) {
  ggplot(df, aes(x = xvar, y = yvar)) +
    geom_point(alpha = 0.6, color = "blue") +
    geom_smooth(method = "lm", color = "red", se = FALSE) +
    labs(title = paste(ylabel, "vs", xlabel), x = xlabel, y = ylabel) +
    theme_minimal()
}

# Generate individual scatterplots
scatterplot_plot(df$Prevelance.of.Undernourishment, df$Life.Expectancy.World.Bank,
                 "Prevalence of Undernourishment", "Life Expectancy (World Bank)")
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 702 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 702 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



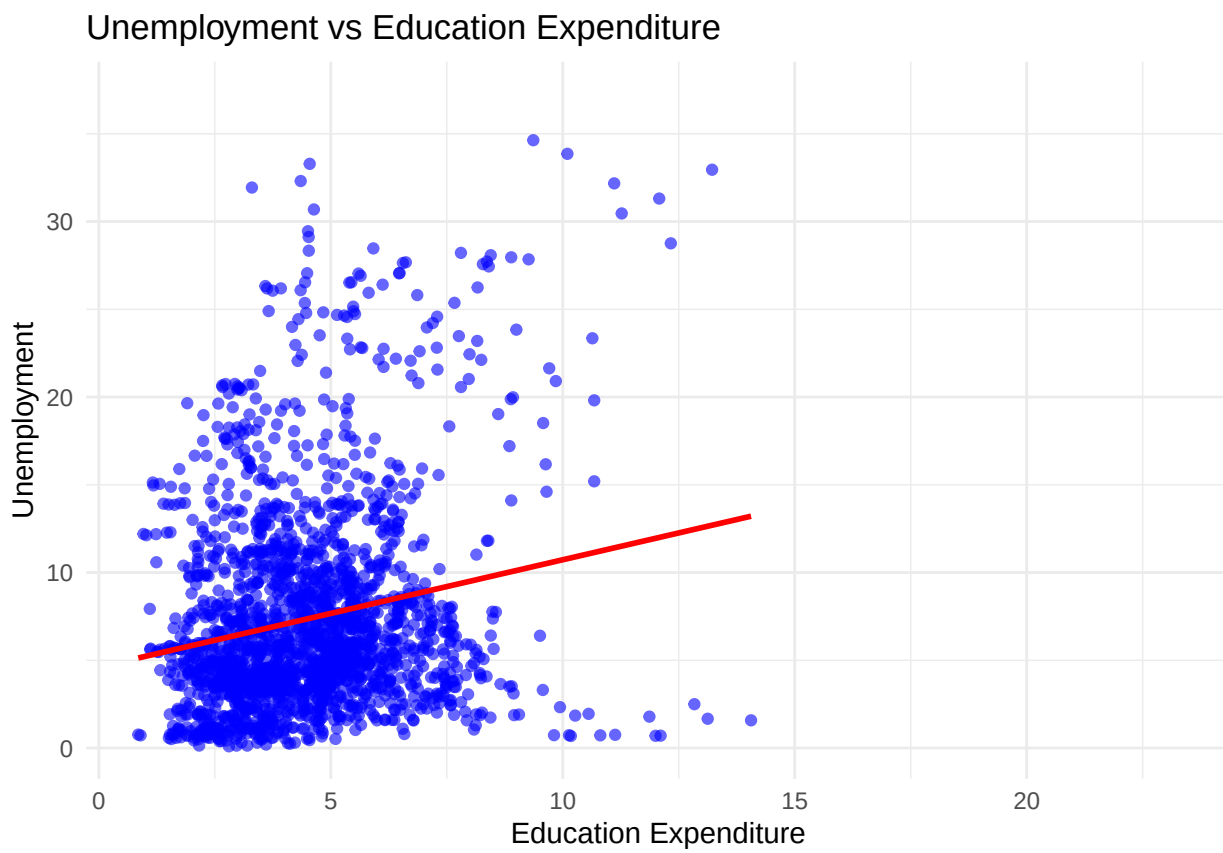

```r
# This scatterplot examines the relationship between undernourishment prevalence and life expectancy.
# The red regression line indicates the trend-potentially showing that higher undernourishment
# correlates with lower life expectancy.

scatterplot_plot(df$Health.Expenditure.., df$Unemployment,
                 "Health Expenditure", "Unemployment")
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 408 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 408 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# This scatterplot investigates whether health expenditure influences unemployment rates.
# A downward-sloping red line might suggest that higher health spending is associated with lower unempl

scatterplot_plot(df$Education.Expenditure.., df$Unemployment,
                 "Education Expenditure", "Unemployment")
```

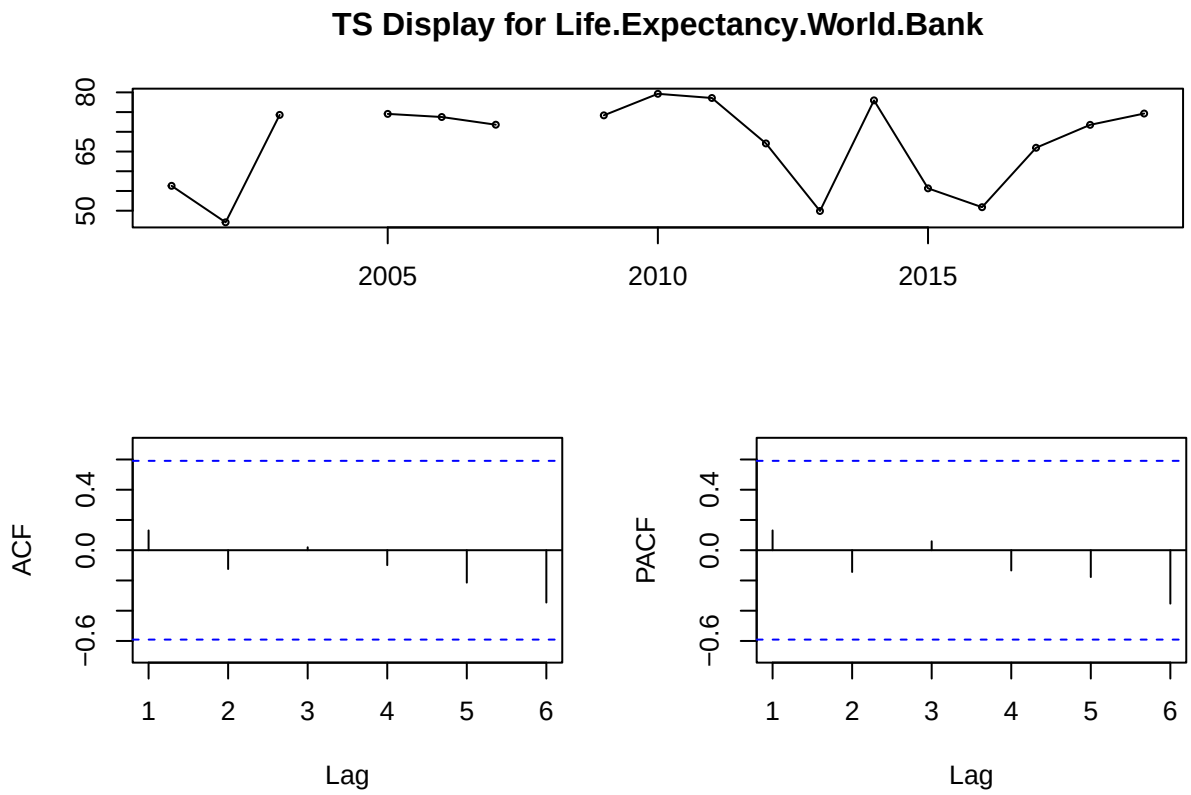
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 1206 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 1206 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



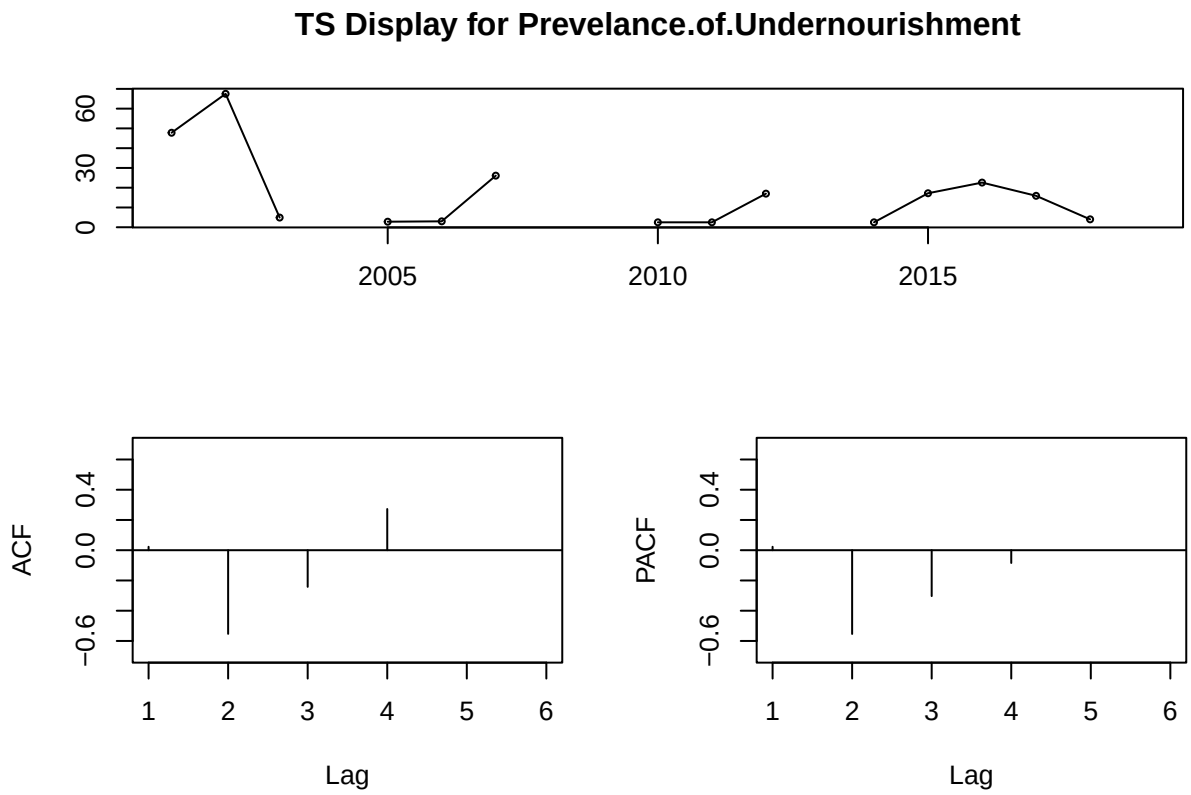

```
# This scatterplot explores the impact of education expenditure on unemployment.
# A negative correlation could indicate that higher education spending leads to lower unemployment rate
```

```
# Select relevant variables
selected_vars <- c("Life.Expectancy.World.Bank", "Prevelance.of.Undernourishment",
                   "Health.Expenditure..", "Unemployment", "Education.Expenditure..")

# Convert each selected variable to a time series object (assuming 'Year' exists in dataset)
for (var in selected_vars) {
  ts_data <- ts(df[[var]], start = min(df$Year), end = max(df$Year), frequency = 1)

  # Generate tsdisplay plot
  print(paste("Time Series Display for:", var))
  tsdisplay(ts_data, main = paste("TS Display for", var))
}
```

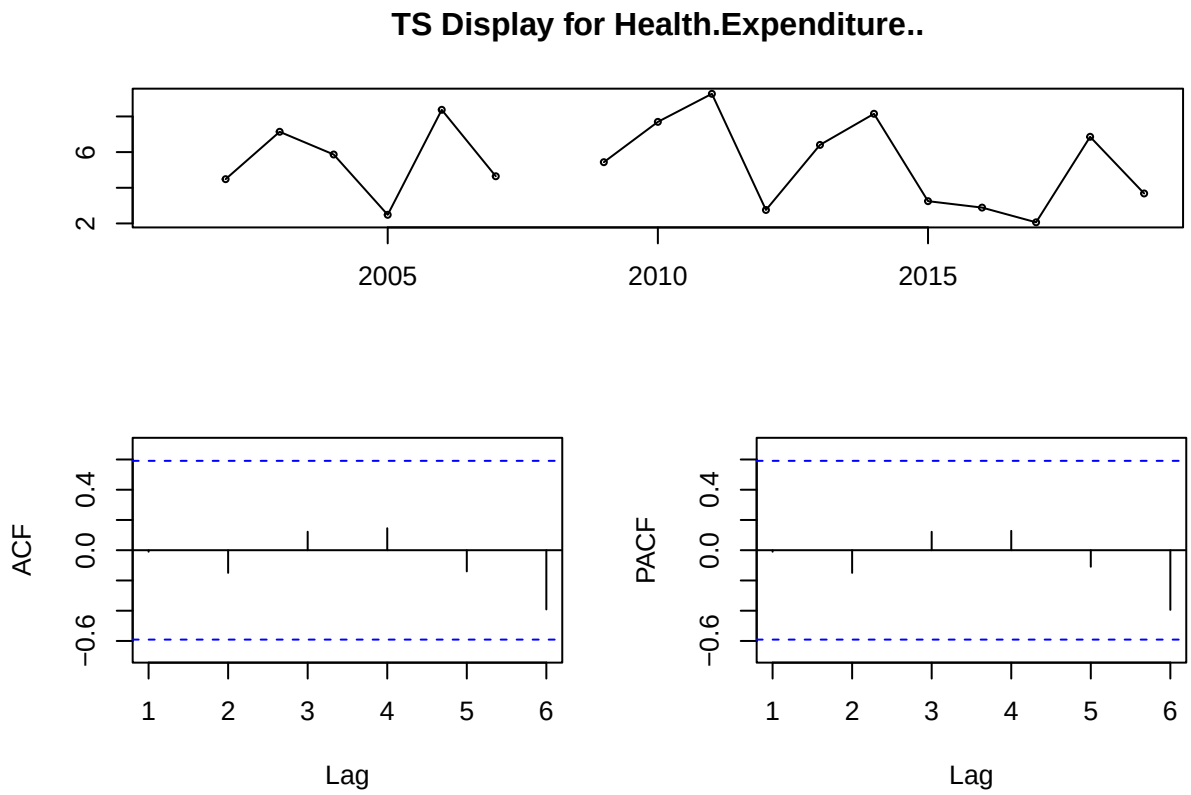
```
## [1] "Time Series Display for: Life.Expectancy.World.Bank"
```



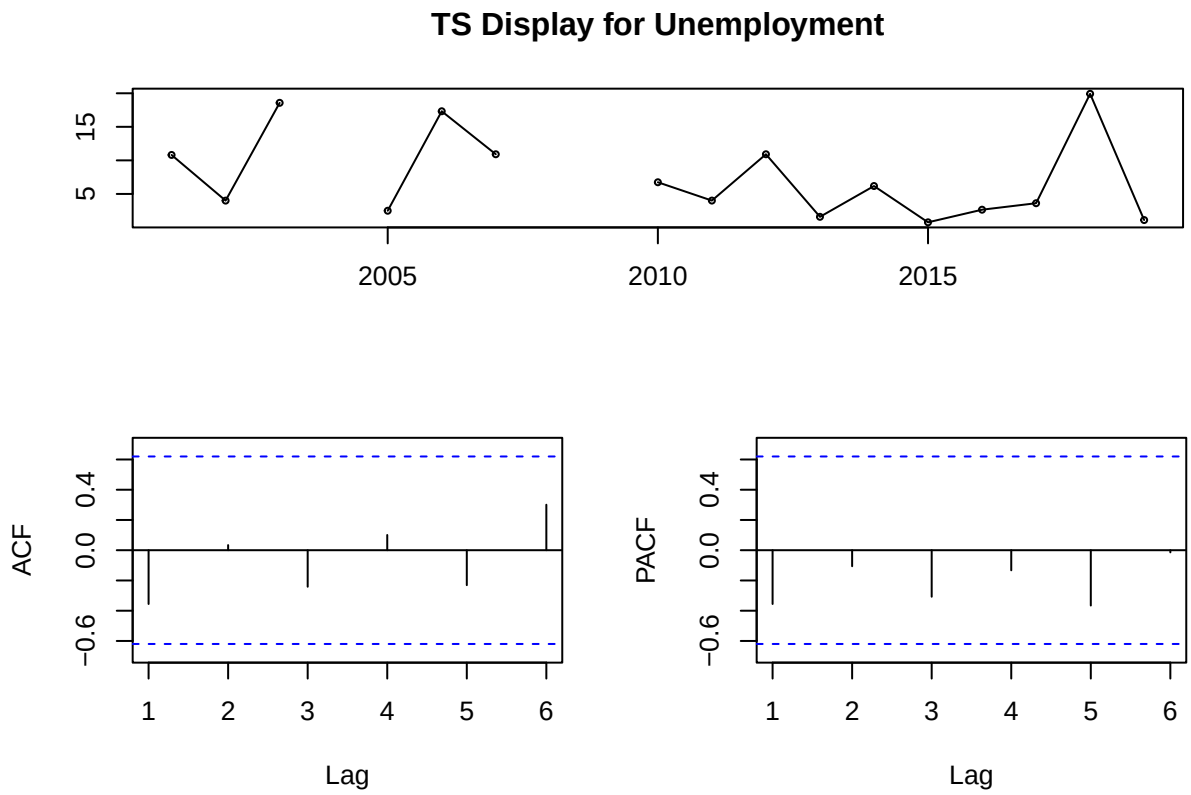

```
## [1] "Time Series Display for: Prevelance.of.Undernourishment"
```



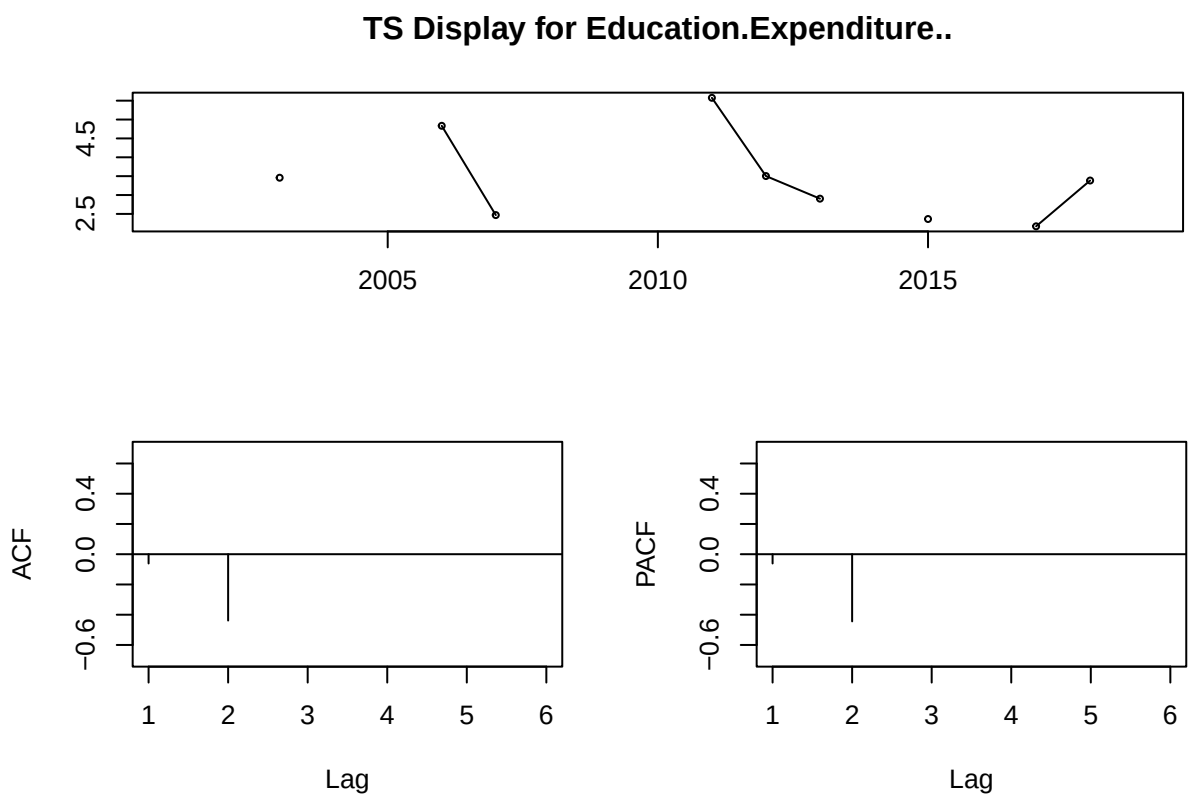

```
## [1] "Time Series Display for: Health.Expenditure.."
```




```
## [1] "Time Series Display for: Unemployment"
```




```
## [1] "Time Series Display for: Education.Expenditure.."
```



Interpretation

Life Expectancy:

- The graphs show non-stationary,explosive trends and unusual behavior for this variable.

Prevalence of Undernourishment:

- Downward trend in the TS plot indicates that the series is non-stationary. The slow decay in both ACF and PACF also indicate that the series is non-stationary.

Health Expenditure:

- TS plot indicates that the series is stationary. But, ACF and PACF exhibit unusual behavior.(Requires differencing)

Unemployment:

- TS plot indicates that the series is ono-stationary. But, ACF and PACF exhibit unusual behavior. (Requires differencing)

Education Expenditure:

- TS plot indicates that the series is non-stationary. ACF and PACF suggest a MA

```r
# Load necessary libraries
library(forecast)
library(tseries)
library(caret)

# Load data
df <- read.csv("life.csv", stringsAsFactors = FALSE)
```

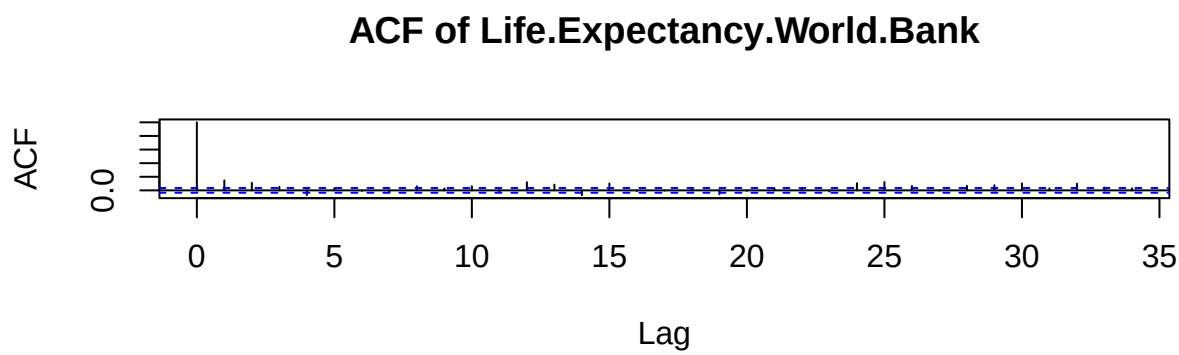
```r
# Convert Year column to numeric (if available)
if ("Year" %in% colnames(df)) {
  df$Year <- as.numeric(df$Year)
}

# Define selected variables
selected_vars <- c("Life.Expectancy.World.Bank", "Prevelance.of.Undernourishment",
                   "Health.Expenditure..", "Unemployment")

# Loop through each variable and fit AR models
for (var in selected_vars) {

  if (var %in% colnames(df)) {

    cat("\n==============================\n")
    cat("Processing:", var, "\n")
    cat("==============================\n")

    # Convert column to numeric and remove NA values
    df[[var]] <- as.numeric(df[[var]])
    ts_data <- ts(na.omit(df[[var]]), start = min(df$Year, na.rm = TRUE), frequency = 1)

    # Ensure the series has enough data points
    if (length(ts_data) < 10) {
      cat("Skipping", var, "- Not enough data points for AR modeling.\n")
      next
    }

    # Check stationarity using ADF test
    adf_test <- tryCatch({
      adf.test(ts_data, alternative = "stationary")
    }, error = function(e) {
      cat("ADF Test failed for", var, "- Skipping differencing.\n")
      return(NULL)
    })

    if (!is.null(adf_test) && adf_test$p.value > 0.05) {
      cat("Series is non-stationary, applying differencing...\n")
      ts_data <- diff(ts_data)
    }

    # ACF & PACF plots
    par(mfrow=c(2,1))
    acf(ts_data, main=paste("ACF of", var))
    pacf(ts_data, main=paste("PACF of", var))
    par(mfrow=c(1,1))

    # Fit AR models
    ar_model_1 <- arima(ts_data, order = c(1,0,0))
    ar_model_2 <- arima(ts_data, order = c(2,0,0))

    # Print model summaries
    cat("\nAR(1) Model Summary:\n")
```

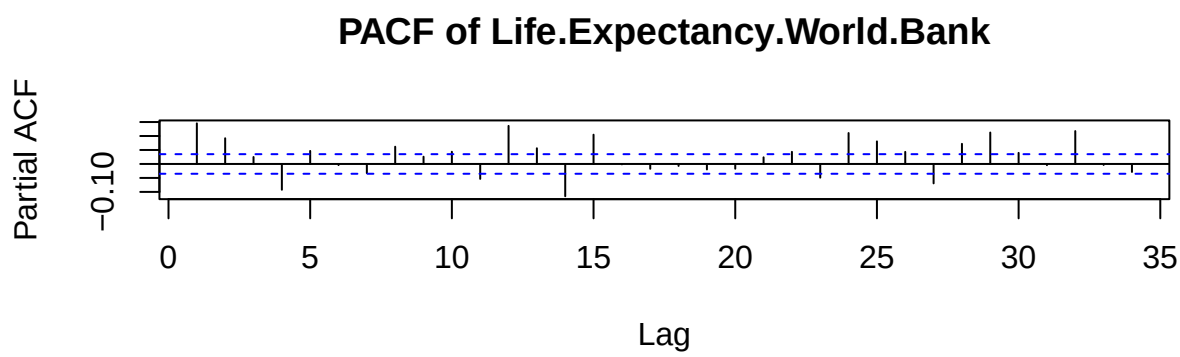
```r
print(ar_model_1)

cat("\nAR(2) Model Summary:\n")
print(ar_model_2)

# Model Selection using AIC and BIC
cat("\nAIC and BIC:\n")
cat("AR(1) - AIC:", AIC(ar_model_1), " BIC:", BIC(ar_model_1), "\n")
cat("AR(2) - AIC:", AIC(ar_model_2), " BIC:", BIC(ar_model_2), "\n")

# Train-test split (2/3 training, 1/3 testing)
train_size <- round(2/3 * length(ts_data))

if (train_size < 5) {
  cat("Skipping", var, "- Not enough data for train-test split.\n")
  next
}

train_data <- ts_data[1:train_size]
test_data <- ts_data[(train_size+1):length(ts_data)]

# Fit AR models on training data
ar_model_1_train <- arima(train_data, order = c(1,0,0))
ar_model_2_train <- arima(train_data, order = c(2,0,0))

# Forecast on test data
forecast_1 <- forecast(ar_model_1_train, h = length(test_data))
forecast_2 <- forecast(ar_model_2_train, h = length(test_data))

# Compute RMSE safely
rmse <- function(actual, predicted) {
  if (length(actual) != length(predicted) || any(is.na(predicted))) {
    return(NA) # Return NA if lengths don't match or there are NA values
  }
  sqrt(mean((actual - predicted)^2, na.rm = TRUE))
}

rmse_train_1 <- rmse(train_data, fitted(ar_model_1_train))
rmse_test_1 <- rmse(test_data, forecast_1$mean)

rmse_train_2 <- rmse(train_data, fitted(ar_model_2_train))
rmse_test_2 <- rmse(test_data, forecast_2$mean)

cat("\nRMSE:\n")
cat("AR(1) - Train:", rmse_train_1, " Test:", rmse_test_1, "\n")
cat("AR(2) - Train:", rmse_train_2, " Test:", rmse_test_2, "\n")

# Compare residuals (ACF and PACF)
if (!any(is.na(resid(ar_model_1_train)))) {
  par(mfrow=c(2,1))
  acf(resid(ar_model_1_train), main=paste("Residuals ACF - AR(1)", var))
  pacf(resid(ar_model_1_train), main=paste("Residuals PACF - AR(1)", var))
  par(mfrow=c(1,1))
```

```r
    }

    if (!any(is.na(resid(ar_model_2_train)))) {
      par(mfrow=c(2,1))
      acf(resid(ar_model_2_train), main=paste("Residuals ACF - AR(2)", var))
      pacf(resid(ar_model_2_train), main=paste("Residuals PACF - AR(2)", var))
      par(mfrow=c(1,1))
    }

    # Generate 10-step-ahead forecast for the best model
    if (AIC(ar_model_1) < AIC(ar_model_2)) {
      best_model <- ar_model_1_train
    } else {
      best_model <- ar_model_2_train
    }

    best_forecast <- forecast(best_model, h = 10)

    # Plot forecast
    plot(best_forecast, main = paste("10-Step Forecast for", var), ylab = var, xlab = "Time")
  }
}
```

```
##
## ==============================
## Processing: Life.Expectancy.World.Bank
## ==============================
```

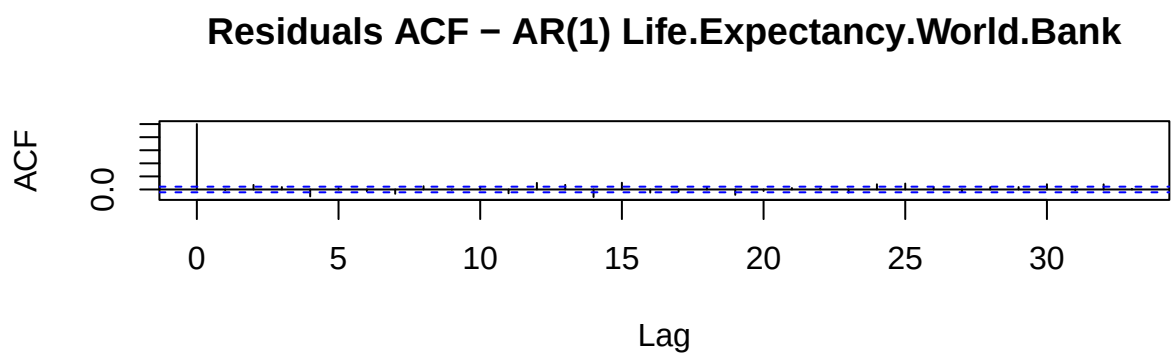
```
## Warning in adf.test(ts_data, alternative = "stationary"): p-value smaller than
## printed p-value
```

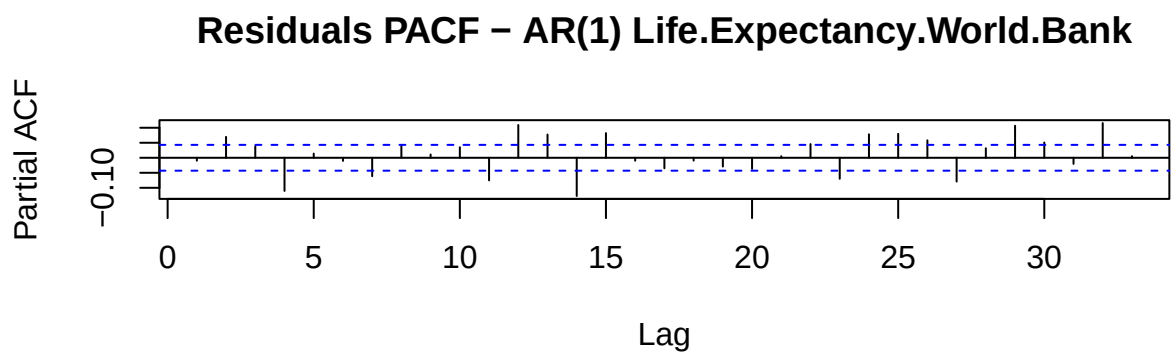


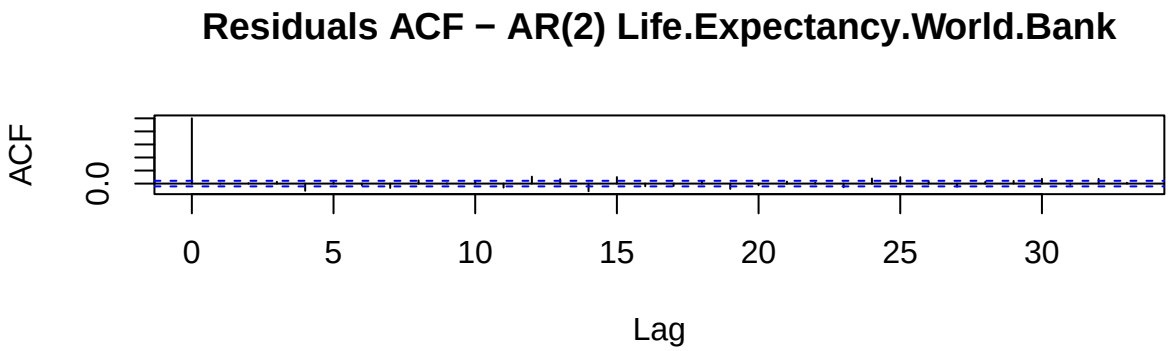


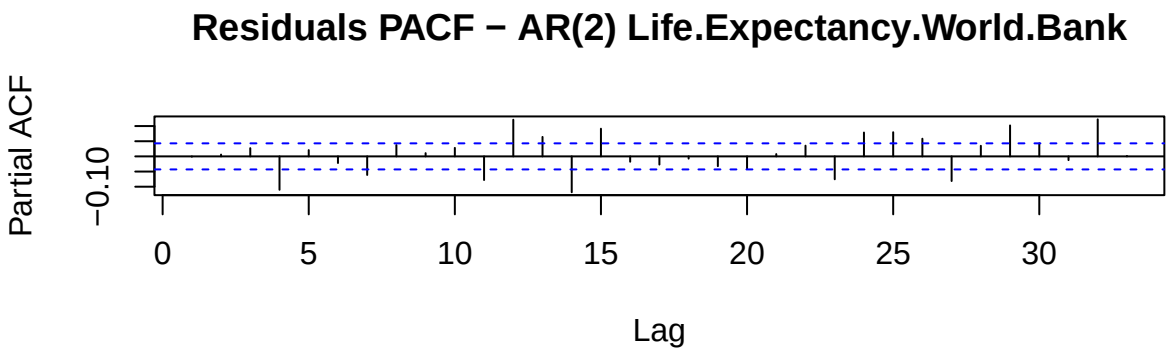

```
##
## AR(1) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(1, 0, 0))
##
## Coefficients:
##          ar1  intercept
##       0.1454    69.7487
## s.e.  0.0177     0.1950
##
## sigma^2 estimated as 86.61:  log likelihood = -11379.69,  aic = 22765.38
##
## AR(2) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(2, 0, 0))
##
## Coefficients:
##          ar1     ar2  intercept
##       0.1321  0.0923    69.7495
## s.e.  0.0178  0.0178     0.2140
##
## sigma^2 estimated as 85.88:  log likelihood = -11366.36,  aic = 22740.73
##
## AIC and BIC:
```

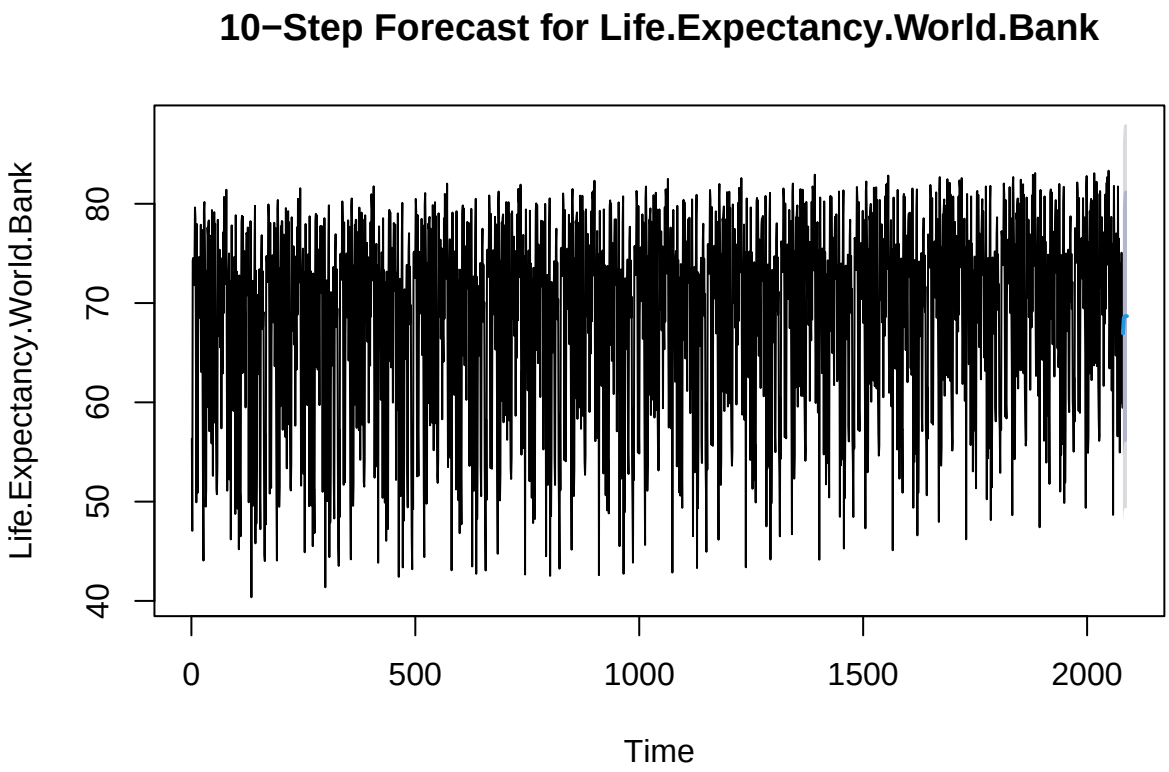
```
## AR(1) - AIC: 22765.38  BIC: 22783.51
## AR(2) - AIC: 22740.73  BIC: 22764.91
##
## RMSE:
## AR(1) - Train: 9.770189  Test: 8.630221
## AR(2) - Train: 9.744128  Test: 8.632058
```



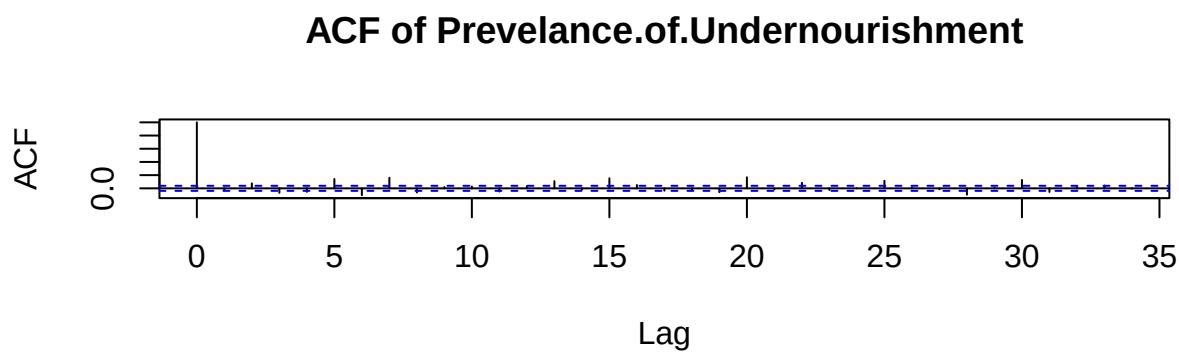


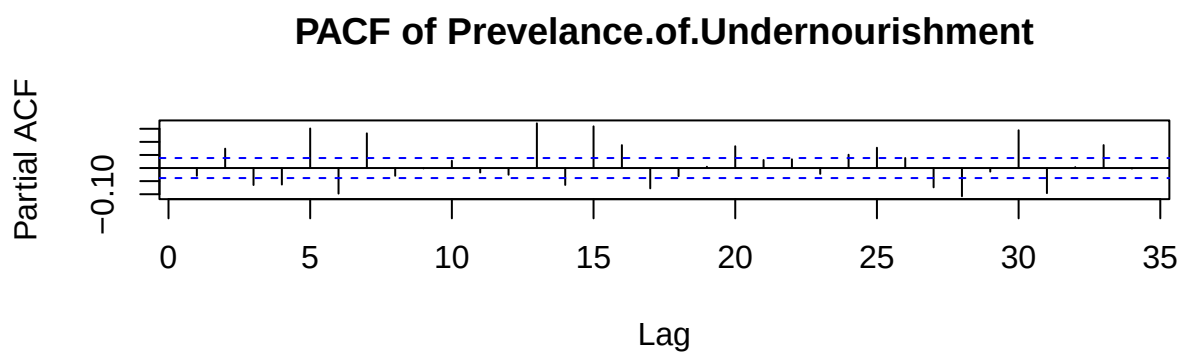

Residuals ACF − AR(2) Life.Expectancy.World.Bank
ACF
0.0
0
5
10
15
20
25
30
Lag


Residuals PACF − AR(2) Life.Expectancy.World.Bank
Partial ACF
−0.10
0
5
10
15
20
25
30
Lag



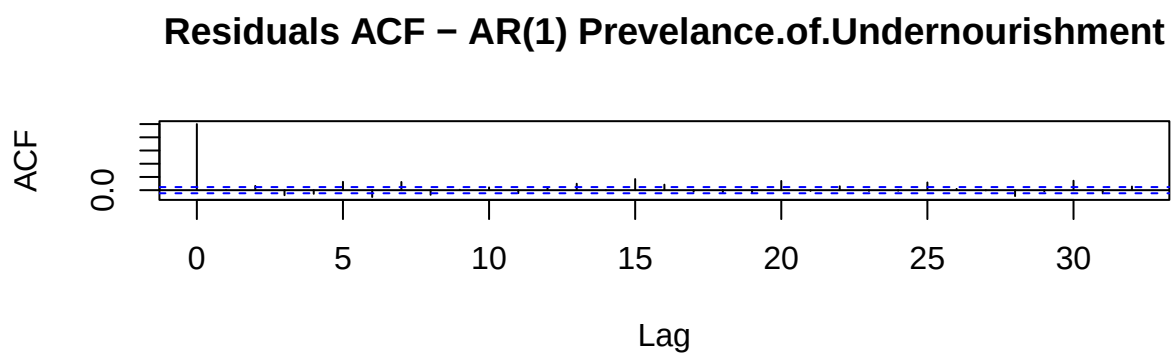

```
##
## ==============================
## Processing: Prevelance.of.Undernourishment
## ==============================

## Warning in adf.test(ts_data, alternative = "stationary"): p-value smaller than
## printed p-value
```

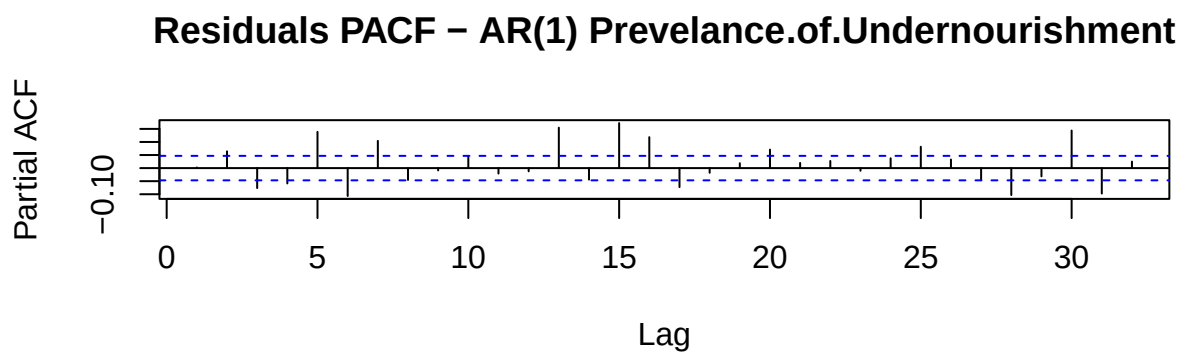


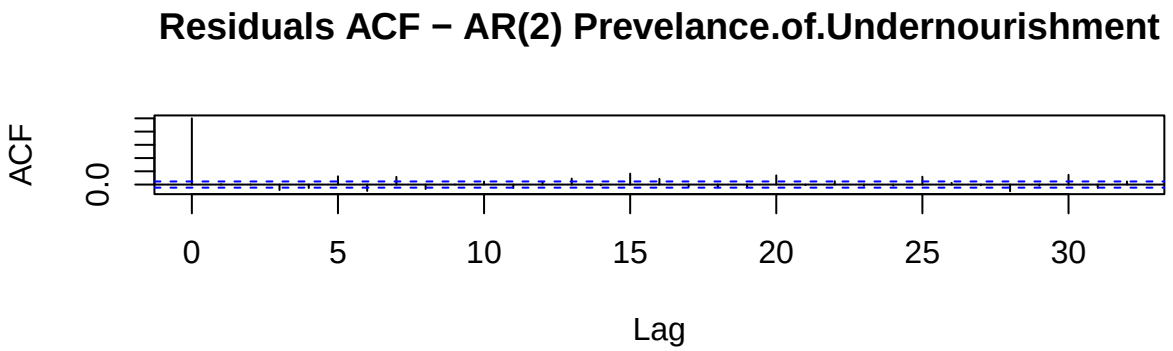


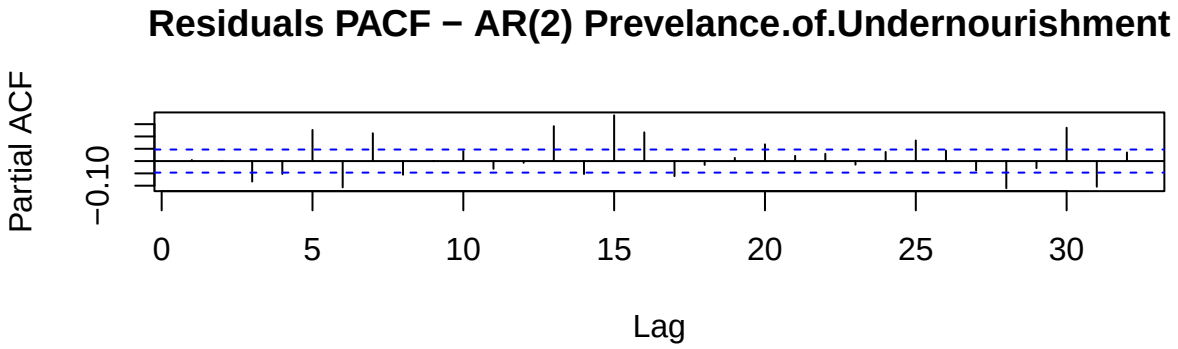

```
##
## AR(1) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(1, 0, 0))
##
## Coefficients:
##           ar1  intercept
##       -0.0296    10.6635
## s.e.   0.0196     0.2139
##
## sigma^2 estimated as 127.2:  log likelihood = -10073.37,  aic = 20152.74
##
## AR(2) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(2, 0, 0))
##
## Coefficients:
##           ar1     ar2  intercept
##       -0.0269  0.0748    10.6666
## s.e.   0.0195  0.0196     0.2307
##
## sigma^2 estimated as 126.5:  log likelihood = -10066.12,  aic = 20140.24
##
## AIC and BIC:
```

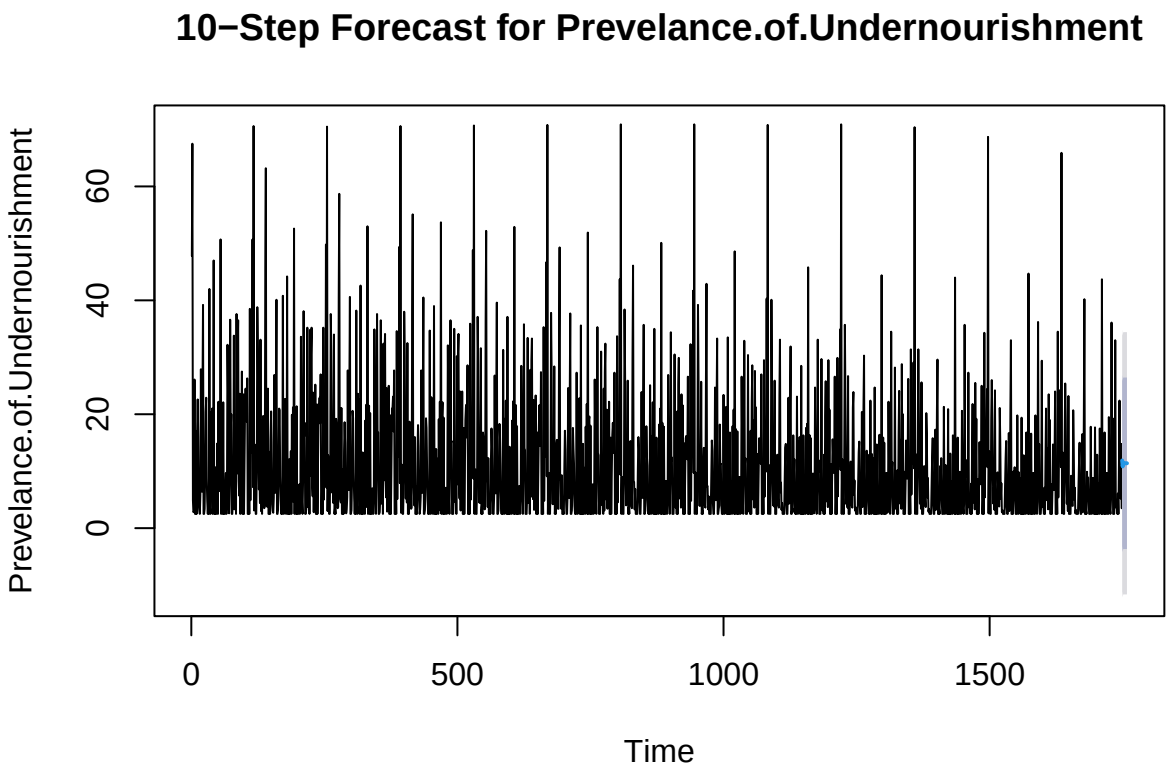
```
## AR(1) - AIC: 20152.74  BIC: 20170.35
## AR(2) - AIC: 20140.24  BIC: 20163.72
##
## RMSE:
## AR(1) - Train: 11.74059  Test: 10.36856
## AR(2) - Train: 11.71409  Test: 10.36897
```



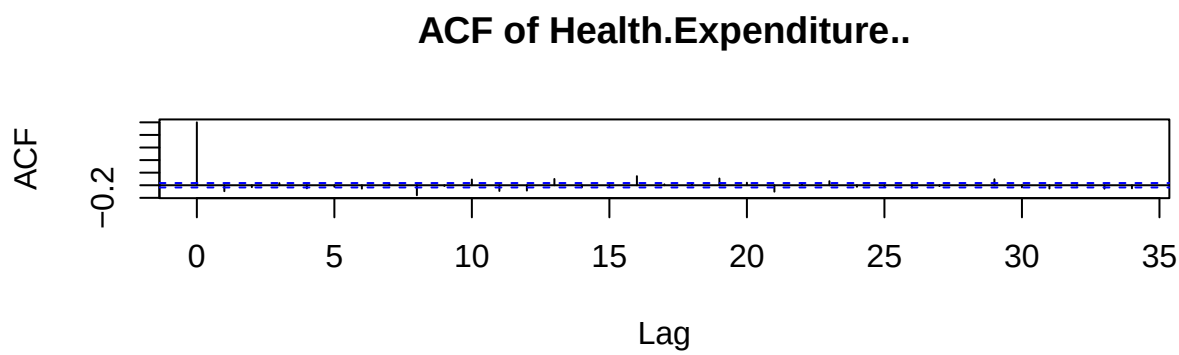


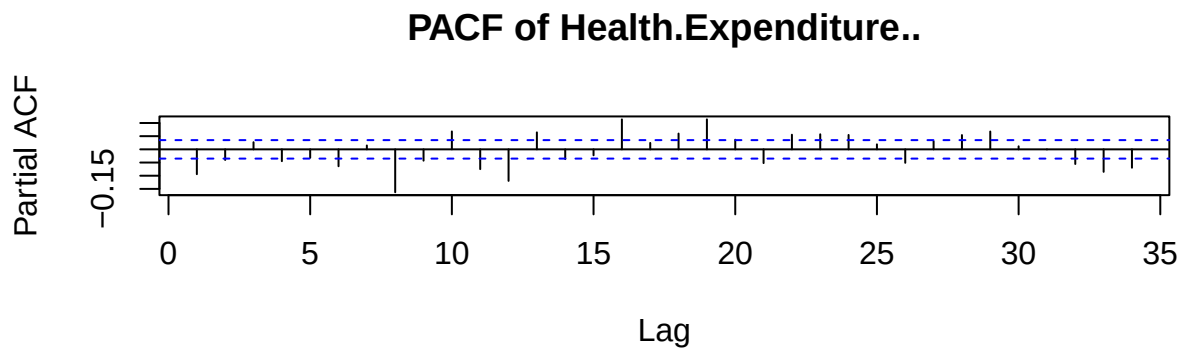

Residuals ACF − AR(2) Prevelance.of.Undernourishment
ACF
0.0
0
5
10
15
20
25
30
Lag


Residuals PACF − AR(2) Prevelance.of.Undernourishment
Partial ACF
−0.10
0
5
10
15
20
25
30
Lag



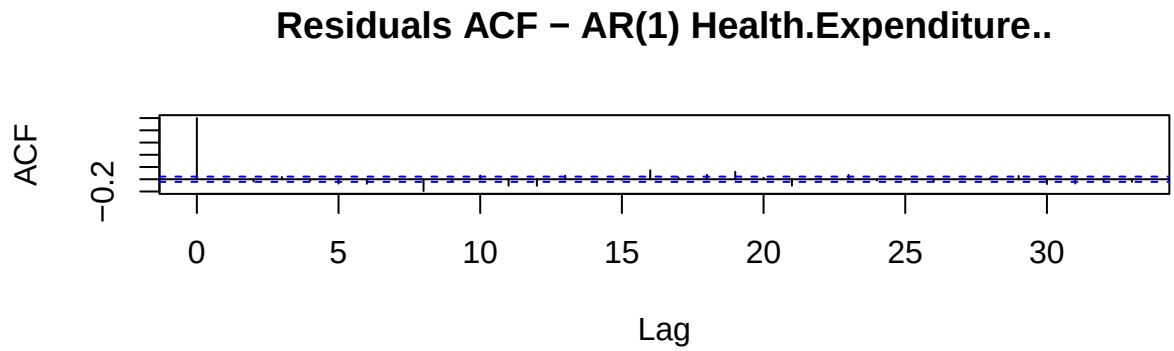

```
##
## ==============================
## Processing: Health.Expenditure..
## ==============================

## Warning in adf.test(ts_data, alternative = "stationary"): p-value smaller than
## printed p-value
```

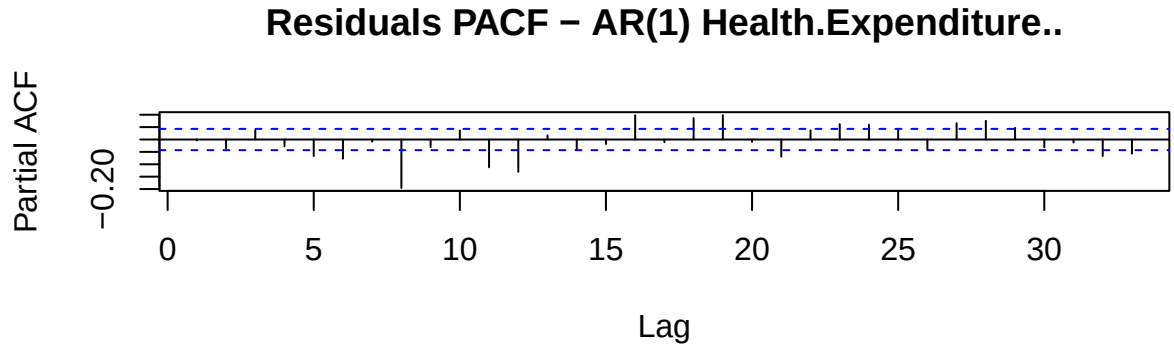


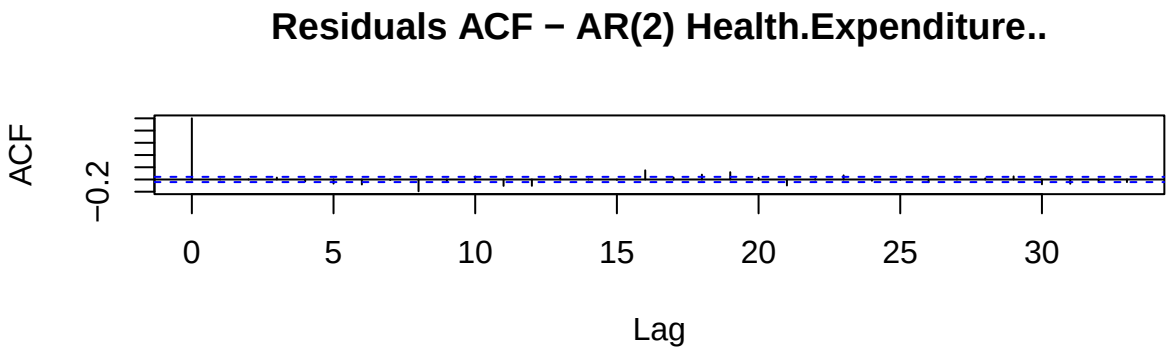


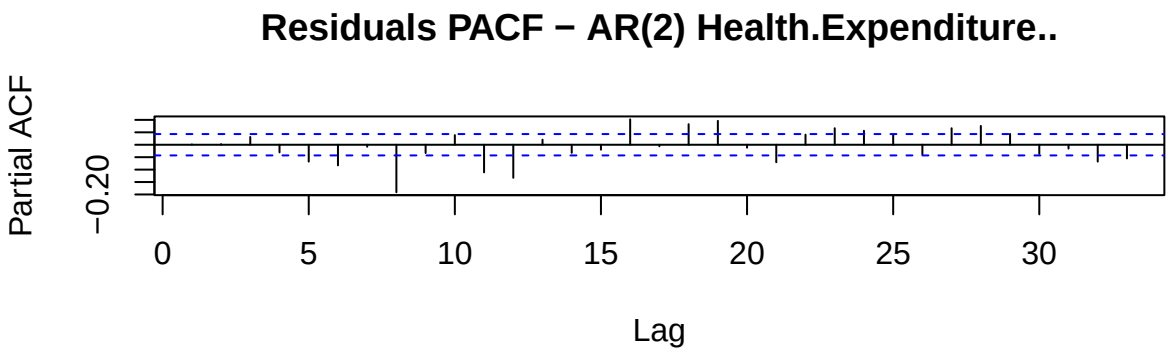

```
##
## AR(1) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(1, 0, 0))
##
## Coefficients:
##           ar1  intercept
##       -0.0943     6.3641
## s.e.   0.0178     0.0463
##
## sigma^2 estimated as 8.007:  log likelihood = -7687.21,  aic = 15380.42
##
## AR(2) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(2, 0, 0))
##
## Coefficients:
##           ar1      ar2  intercept
##       -0.0981  -0.0403     6.3641
## s.e.   0.0179   0.0179     0.0444
##
## sigma^2 estimated as 7.994:  log likelihood = -7684.67,  aic = 15377.35
##
## AIC and BIC:
```

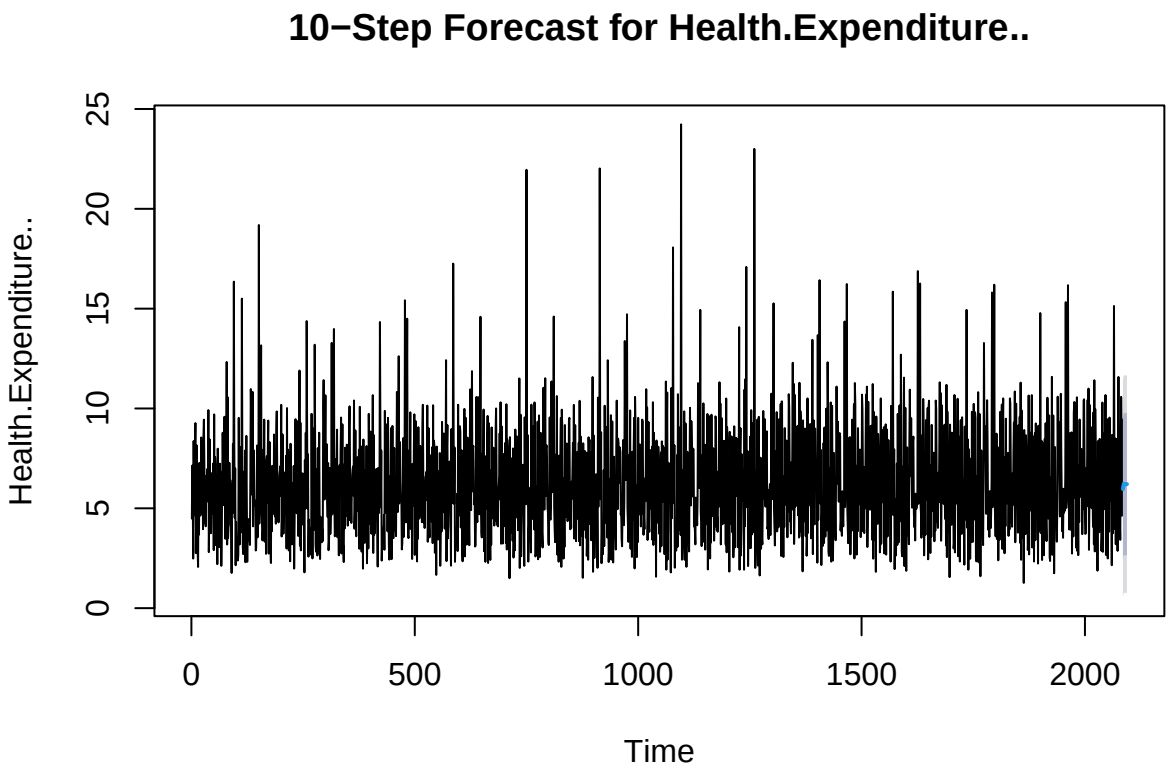
```
## AR(1) - AIC: 15380.42  BIC: 15398.56
## AR(2) - AIC: 15377.35  BIC: 15401.54
##
## RMSE:
## AR(1) - Train: 2.766236  Test: 2.979138
## AR(2) - Train: 2.764062  Test: 2.97897
```



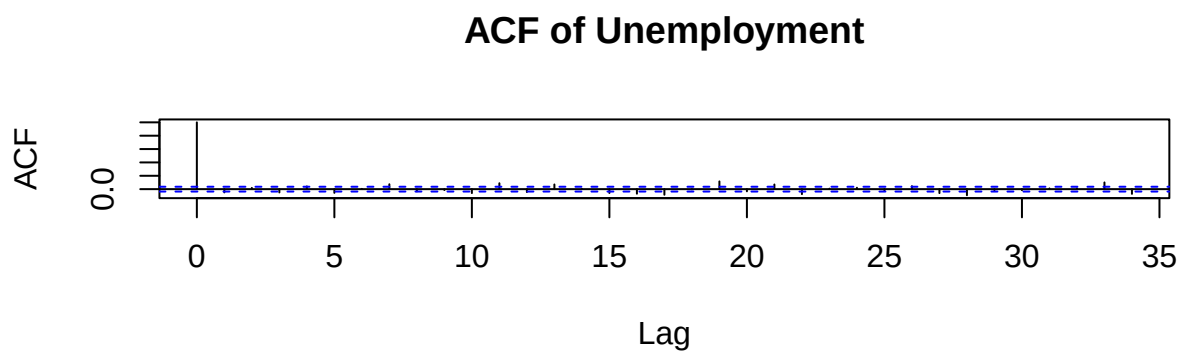


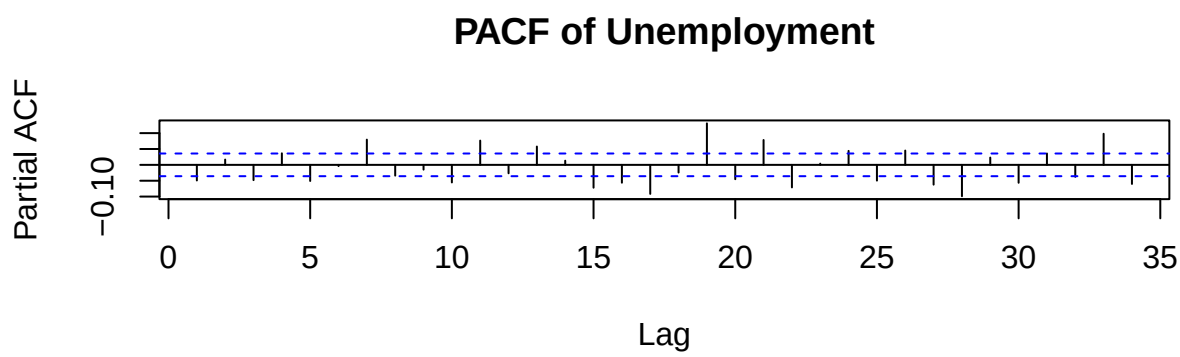

Residuals ACF − AR(2) Health.Expenditure..
ACF
−0.2
0
5
10
15
20
25
30
Lag


Residuals PACF − AR(2) Health.Expenditure..
Partial ACF
−0.20
0
5
10
15
20
25
30
Lag



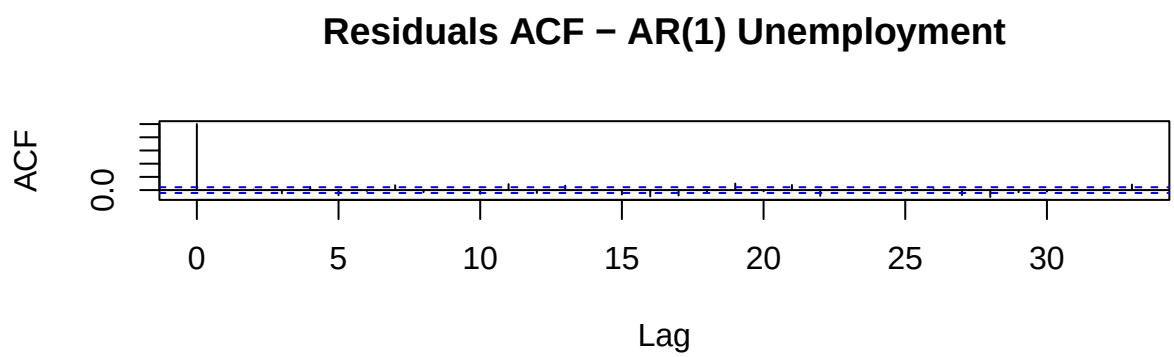

```
##
## ==============================
## Processing: Unemployment
## ==============================

## Warning in adf.test(ts_data, alternative = "stationary"): p-value smaller than
## printed p-value
```

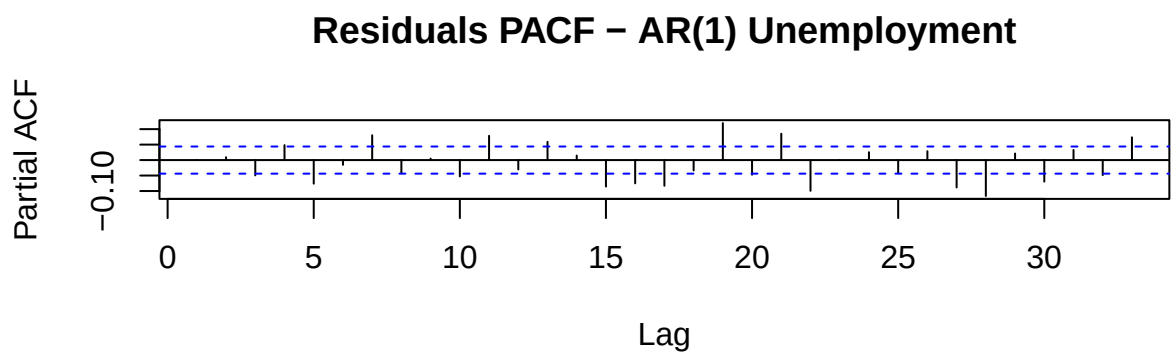


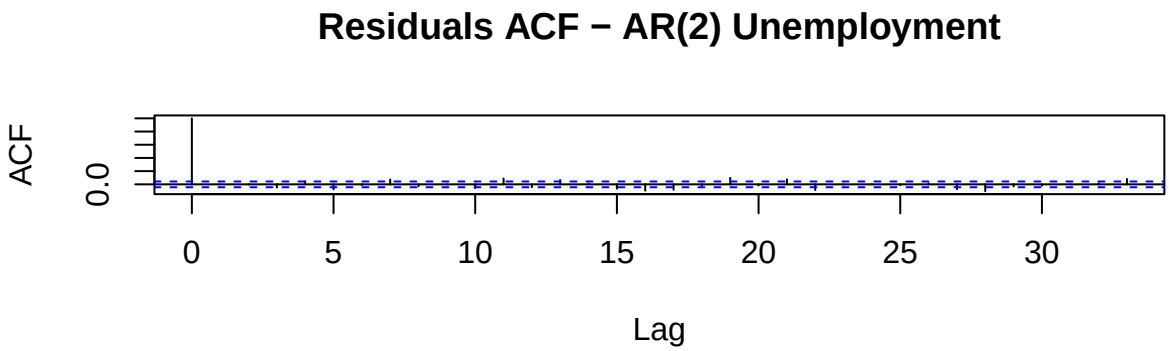


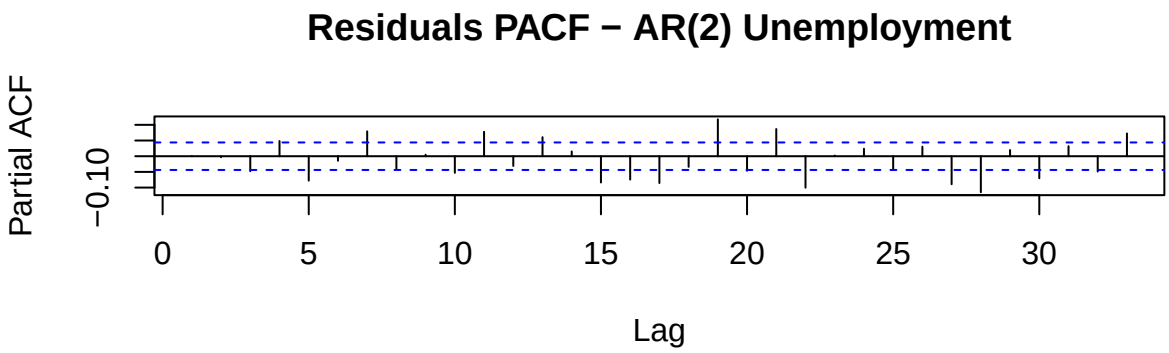

```
##
## AR(1) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(1, 0, 0))
##
## Coefficients:
##           ar1  intercept
##       -0.0492     7.8907
## s.e.   0.0182     0.1089
##
## sigma^2 estimated as 39.21:  log likelihood = -9766.91,  aic = 19539.82
##
## AR(2) Model Summary:
##
## Call:
## arima(x = ts_data, order = c(2, 0, 0))
##
## Coefficients:
##           ar1     ar2  intercept
##       -0.0484  0.0167     7.8908
## s.e.   0.0182  0.0182     0.1108
##
## sigma^2 estimated as 39.2:  log likelihood = -9766.49,  aic = 19540.98
##
## AIC and BIC:
```

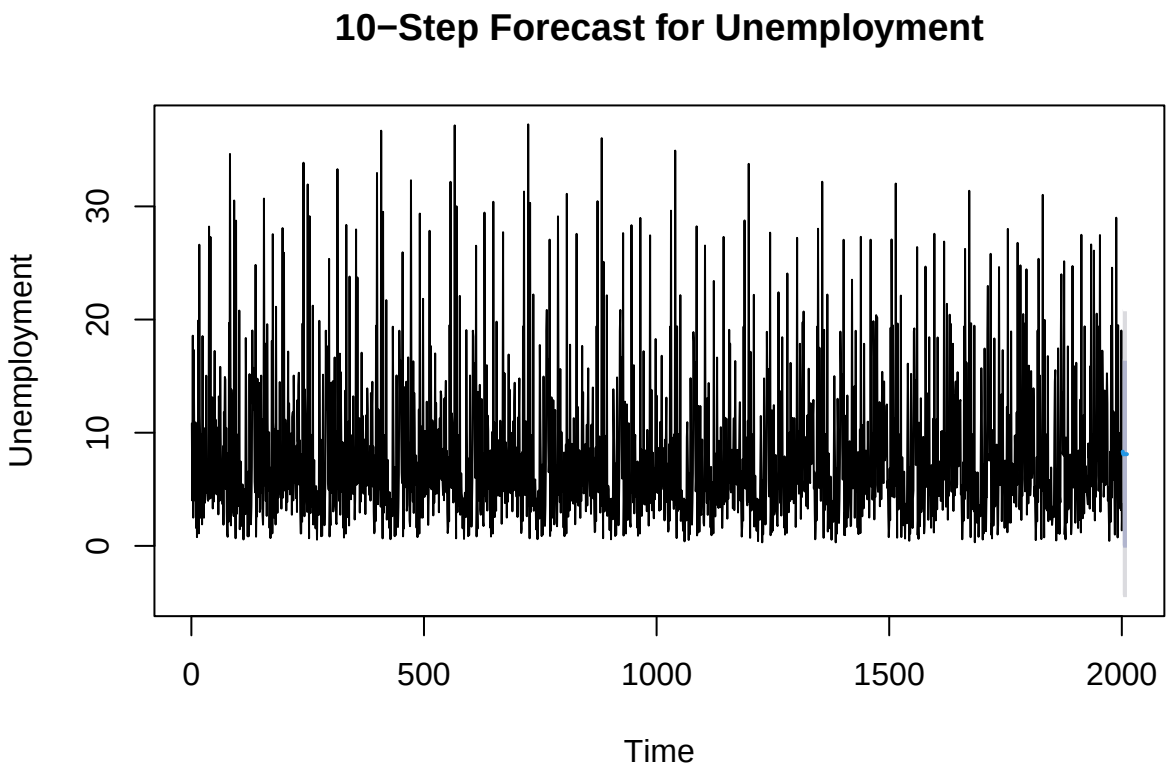
```
## AR(1) - AIC: 19539.82  BIC: 19557.84
## AR(2) - AIC: 19540.98  BIC: 19565.01
##
## RMSE:
## AR(1) - Train: 6.435168  Test: 5.918082
## AR(2) - Train: 6.434693  Test: 5.918064
```






Residuals ACF − AR(2) Unemployment
ACF
0.0
0
5
10
15
20
25
30
Lag


Residuals PACF − AR(2) Unemployment
Partial ACF
−0.10
0
5
10
15
20
25
30
Lag


10−Step Forecast for Unemployment
Unemployment
0
10
20
30
0
500
1000
1500
2000
Time

Interpretation

- AIC & BIC: The AR(2) model has a slightly lower AIC (15377.35 vs. 15380.42), suggesting a better fit.
- RMSE: The RMSE values are very close for AR(1) and AR(2), meaning both models predict similarly.
- Coefficients: The AR(2) model includes an extra lag term (ar2 = -0.0403), but it has a small effect.
- Conclusion: The AR(2) model is slightly better based on AIC, but the difference is minor. Both models fit well.

AR Model for Unemployment:

- AIC & BIC: The AR(1) model has a slightly lower AIC (19539.82 vs. 19540.98), indicating it is marginally better.
- RMSE: The test RMSE is very close for both models, meaning they predict similarly.
- Coefficients: The AR(2) model has an extra lag term (ar2 = 0.0167), but the effect is very weak.

-Conclusion: AR(1) is slightly better due to lower AIC and simpler structure. AR(2) does not add much value.

Final Model Selection:

- Based on AIC, BIC, and RMSE, the best models are:

Life Expectancy: AR(2) is slightly better, but AR(1) is also reasonable. Unemployment: AR(1) is better, as AR(2) does not improve fit significantly

- Forecasts show reasonable future predictions.

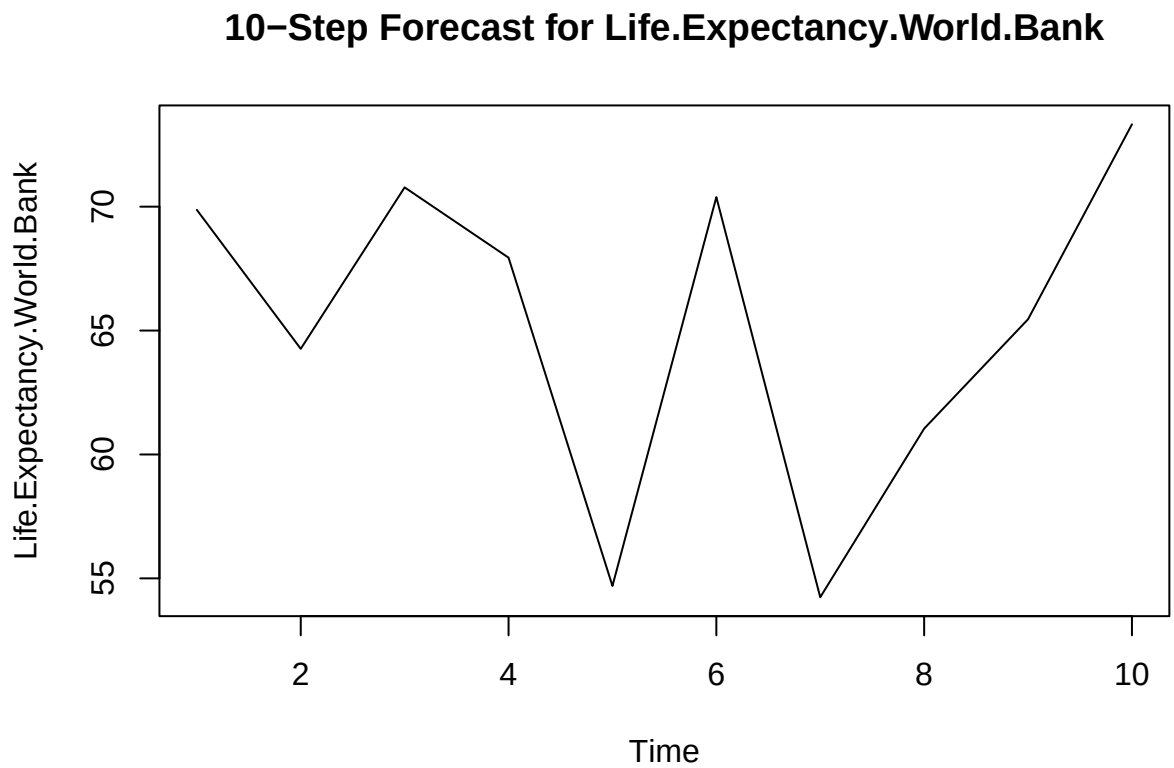
```r
# Load data
df <- read.csv("life.csv", stringsAsFactors = FALSE)

# Convert Year to numeric
if ("Year" %in% colnames(df)) {
  df$Year <- as.numeric(df$Year)
}

# Define response variables and predictors
response_vars <- c("Life.Expectancy.World.Bank", "Unemployment")
predictors <- c("Prevelance.of.Undernourishment", "Health.Expenditure..", "Education.Expenditure..")

# Ensure variables are numeric
df[response_vars] <- lapply(df[response_vars], as.numeric)
df[predictors] <- lapply(df[predictors], as.numeric)

# Remove missing values
df <- na.omit(df)

# Function to fit ARDL models
fit_ARDL <- function(response, predictor) {
  cat("\n==============================\n")
  cat("Processing:", response, "with predictor:", predictor, "\n")
  cat("==============================\n")

  # Convert to time series
```

```r
y <- ts(df[[response]], start = min(df$Year), frequency = 1)
x <- ts(df[[predictor]], start = min(df$Year), frequency = 1)

# Check stationarity and apply differencing if necessary
adf_y <- adf.test(y, alternative = "stationary")
if (adf_y$p.value > 0.05) {
  y <- diff(y)
  cat("Differenced response variable for stationarity.\n")
}

adf_x <- adf.test(x, alternative = "stationary")
if (adf_x$p.value > 0.05) {
  x <- diff(x)
  cat("Differenced predictor variable for stationarity.\n")
}

# Convert time series to a data frame with correct lag structure
df_ardl <- data.frame(
  y = y[-1],
  L_y1 = stats::lag(y, -1)[-1],
  L_x1 = stats::lag(x, -1)[-1]
) %>% na.omit()

# Fit ARDL(1,1) and ARDL(2,2) models
ardl_model_1 <- lm(y ~ L_y1 + L_x1, data = df_ardl)
df_ardl$L_y2 <- stats::lag(y, -2)[-1]
df_ardl$L_x2 <- stats::lag(x, -2)[-1]
df_ardl <- na.omit(df_ardl)
ardl_model_2 <- lm(y ~ L_y2 + L_x2, data = df_ardl)

# Print model summaries
cat("\nARDL(1,1) Model Summary:\n")
print(summary(ardl_model_1))

cat("\nARDL(2,2) Model Summary:\n")
print(summary(ardl_model_2))

# Model Selection using AIC and BIC
aic_1 <- AIC(ardl_model_1)
bic_1 <- BIC(ardl_model_1)
aic_2 <- AIC(ardl_model_2)
bic_2 <- BIC(ardl_model_2)

cat("\nAIC and BIC:\n")
cat("ARDL(1,1) - AIC:", aic_1, " BIC:", bic_1, "\n")
cat("ARDL(2,2) - AIC:", aic_2, " BIC:", bic_2, "\n")

# Train-test split (2/3 training, 1/3 testing)
train_size <- round(2/3 * nrow(df_ardl))
train_data <- df_ardl[1:train_size, ]
test_data <- df_ardl[(train_size+1):nrow(df_ardl), ]

# Fit ARDL models on training data
```

```r
  ardl_train_1 <- lm(y ~ L_y1 + L_x1, data = train_data)
  ardl_train_2 <- lm(y ~ L_y2 + L_x2, data = train_data)

  # Forecast on test data
  forecast_1 <- predict(ardl_train_1, newdata = test_data)
  forecast_2 <- predict(ardl_train_2, newdata = test_data)

  # Debugging: Check model coefficients vs. new_forecast_data
  cat("\nChecking model coefficients:\n")
  print(names(coef(ardl_train_1)))

  # Prepare `newdata` for forecasting
  new_forecast_data <- tail(df_ardl, 10)  # Get last 10 observations

  # Ensure `newdata` matches model structure
  common_columns <- intersect(names(new_forecast_data), names(coef(ardl_train_1)))
  new_forecast_data <- new_forecast_data[, common_columns, drop = FALSE]

  # Debugging: Confirm `new_forecast_data` before applying predict()
  cat("\nFinal `new_forecast_data` structure:\n")
  print(colnames(new_forecast_data))

  # Generate 10-step-ahead forecast using `predict()`
  best_forecast <- predict(ardl_train_1, newdata = new_forecast_data)

  # Plot forecast
  plot.ts(best_forecast, main = paste("10-Step Forecast for", response), ylab = response, xlab = "Time")
}

# Fit ARDL models for Life Expectancy and Unemployment
fit_ARDL("Life.Expectancy.World.Bank", "Health.Expenditure..")
```

```
##
## ==============================
## Processing: Life.Expectancy.World.Bank with predictor: Health.Expenditure..
## ==============================

## Warning in adf.test(y, alternative = "stationary"): p-value smaller than
## printed p-value

## Warning in adf.test(x, alternative = "stationary"): p-value smaller than
## printed p-value

##
## ARDL(1,1) Model Summary:

## Warning in summary.lm(ardl_model_1): essentially perfect fit: summary may be
## unreliable

##
## Call:
## lm(formula = y ~ L_y1 + L_x1, data = df_ardl)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -2.336e-14 -1.234e-16  1.074e-16  2.364e-16  1.562e-15
```

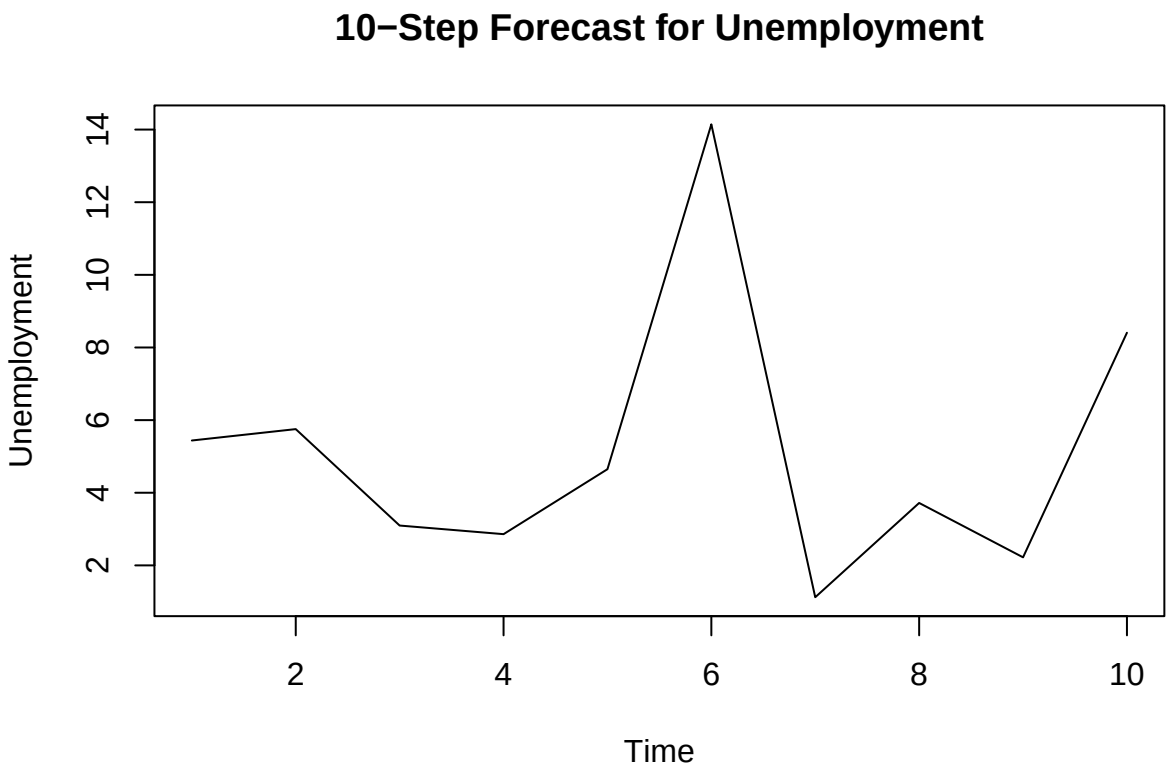
```
##
## Coefficients:
##              Estimate Std. Error   t value Pr(>|t|)
## (Intercept) 1.347e-14  7.550e-16 1.784e+01   <2e-16 ***
## L_y1        1.000e+00  1.106e-17 9.042e+16   <2e-16 ***
## L_x1        5.562e-17  3.312e-17 1.679e+00   0.0942 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.441e-15 on 282 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 4.237e+33 on 2 and 282 DF,  p-value: < 2.2e-16
##
##
## ARDL(2,2) Model Summary:

## Warning in summary.lm(ardl_model_2): essentially perfect fit: summary may be
## unreliable

##
## Call:
## lm(formula = y ~ L_y2 + L_x2, data = df_ardl)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -2.336e-14 -1.234e-16  1.074e-16  2.364e-16  1.562e-15
##
## Coefficients:
##              Estimate Std. Error   t value Pr(>|t|)
## (Intercept) 1.347e-14  7.550e-16 1.784e+01   <2e-16 ***
## L_y2        1.000e+00  1.106e-17 9.042e+16   <2e-16 ***
## L_x2        5.562e-17  3.312e-17 1.679e+00   0.0942 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.441e-15 on 282 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 4.237e+33 on 2 and 282 DF,  p-value: < 2.2e-16
##
##
## AIC and BIC:
## ARDL(1,1) - AIC: -18665.18  BIC: -18650.57
## ARDL(2,2) - AIC: -18665.18  BIC: -18650.57
##
## Checking model coefficients:
## [1] "(Intercept)" "L_y1"        "L_x1"
##
## Final `new_forecast_data` structure:
## [1] "L_y1" "L_x1"
```




```r
fit_ARDL("Unemployment", "Education.Expenditure..")
```

```
##
## ==============================
## Processing: Unemployment with predictor: Education.Expenditure..
## ==============================

## Warning in adf.test(y, alternative = "stationary"): p-value smaller than
## printed p-value
## Warning in adf.test(y, alternative = "stationary"): p-value smaller than
## printed p-value

##
## ARDL(1,1) Model Summary:

## Warning in summary.lm(ardl_model_1): essentially perfect fit: summary may be
## unreliable

##
## Call:
## lm(formula = y ~ L_y1 + L_x1, data = df_ardl)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -4.857e-14  6.000e-18  1.480e-16  3.290e-16  2.877e-15
##
## Coefficients:
##               Estimate Std. Error    t value Pr(>|t|)
```

```
## (Intercept) -1.684e-15  4.177e-16 -4.031e+00 7.16e-05 ***
## L_y1         1.000e+00  2.889e-17  3.461e+16  < 2e-16 ***
## L_x1         5.878e-17  1.007e-16  5.840e-01     0.56
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.943e-15 on 282 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 7.45e+32 on 2 and 282 DF,  p-value: < 2.2e-16
##
##
## ARDL(2,2) Model Summary:

## Warning in summary.lm(ardl_model_2): essentially perfect fit: summary may be
## unreliable

##
## Call:
## lm(formula = y ~ L_y2 + L_x2, data = df_ardl)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -4.857e-14  6.000e-18  1.480e-16  3.290e-16  2.877e-15
##
## Coefficients:
##               Estimate Std. Error    t value Pr(>|t|)
## (Intercept) -1.684e-15  4.177e-16 -4.031e+00 7.16e-05 ***
## L_y2         1.000e+00  2.889e-17  3.461e+16  < 2e-16 ***
## L_x2         5.878e-17  1.007e-16  5.840e-01     0.56
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.943e-15 on 282 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 7.45e+32 on 2 and 282 DF,  p-value: < 2.2e-16
##
##
## AIC and BIC:
## ARDL(1,1) - AIC: -18258.09  BIC: -18243.48
## ARDL(2,2) - AIC: -18258.09  BIC: -18243.48
##
## Checking model coefficients:
## [1] "(Intercept)" "L_y1"        "L_x1"
##
## Final `new_forecast_data` structure:
## [1] "L_y1" "L_x1"
```


10−Step Forecast for Unemployment
Unemployment
2
4
6
8
10
12
14
2
4
6
8
10
Time

ARDL Model Interpretation

** Perfect Fit Warning**

Warning: Essentially perfect fit: summary may be unreliable

- The model shows **perfect collinearity**, meaning the predictor (`L_x1`, `L_x2`) and response (`y`) are **identical or linearly dependent**.
- **Multiple R-squared = 1** suggests that the model explains **100%** of the variation in `y`, which is **highly unusual** in real-world data.

** Possible Causes**

- The series might be **non-stationary** (not differenced properly).
- `L_x1` and `L_x2` could be **highly correlated** with `y`, creating redundancy.
- Intercept is nearly 0, meaning the model expects y to be fully explained by lags. L_y1 coefficient = 1.000, indicating an exact autoregressive relationship (likely non-stationarity). L_x1 is close to 0, with p-value = 0.0942 (not statistically significant).
- AIC/BIC values are extremely low, but given the perfect fit warnings, these may be unreliable.

```r
# Load required libraries
library(vars)
library(forecast)
library(tseries)
library(lmtest)
library(urca)
library(ggplot2)
library(gridExtra)
library(strucchange)



# Define the variables
y1t <- as.numeric(df$Life.Expectancy.World.Bank)  # First time series variable
y2t <- as.numeric(df$Unemployment)               # Second time series variable

# Convert to time series
y1t_ts <- ts(y1t, start = min(df$Year), frequency = 1)
y2t_ts <- ts(y2t, start = min(df$Year), frequency = 1)

# Combine into a data frame for VAR model
var_data <- data.frame(y1t = y1t_ts, y2t = y2t_ts)

# Check for stationarity using ADF test
cat("\nADF Test for y1t:\n")
```

```
##
## ADF Test for y1t:
```

```r
print(adf.test(y1t_ts))
```

```
## Warning in adf.test(y1t_ts): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
```

```
## data:  y1t_ts
## Dickey-Fuller = -8.0677, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

```r
cat("\nADF Test for y2t:\n")
```

```
##
## ADF Test for y2t:
```

```r
print(adf.test(y2t_ts))
```

```
## Warning in adf.test(y2t_ts): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  y2t_ts
## Dickey-Fuller = -7.1852, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

```r
# Difference the series if needed
if (adf.test(y1t_ts)$p.value > 0.05) {
  y1t_ts <- diff(y1t_ts)
  cat("Differenced y1t to ensure stationarity.\n")
}
```

```
## Warning in adf.test(y1t_ts): p-value smaller than printed p-value
```

```r
if (adf.test(y2t_ts)$p.value > 0.05) {
  y2t_ts <- diff(y2t_ts)
  cat("Differenced y2t to ensure stationarity.\n")
}
```

```
## Warning in adf.test(y2t_ts): p-value smaller than printed p-value
```

```r
# Recreate the dataset after differencing
var_data <- data.frame(y1t = y1t_ts[-1], y2t = y2t_ts[-1])

# Fit the VAR model (automatically selects optimal lag)
lag_selection <- VARselect(var_data, lag.max = 10, type = "const")
opt_lag <- lag_selection$selection["AIC(n)"]  # Choosing based on AIC
var_model <- VAR(var_data, p = opt_lag, type = "const")

cat("\nVAR Model Summary:\n")
```

```
##
## VAR Model Summary:
```

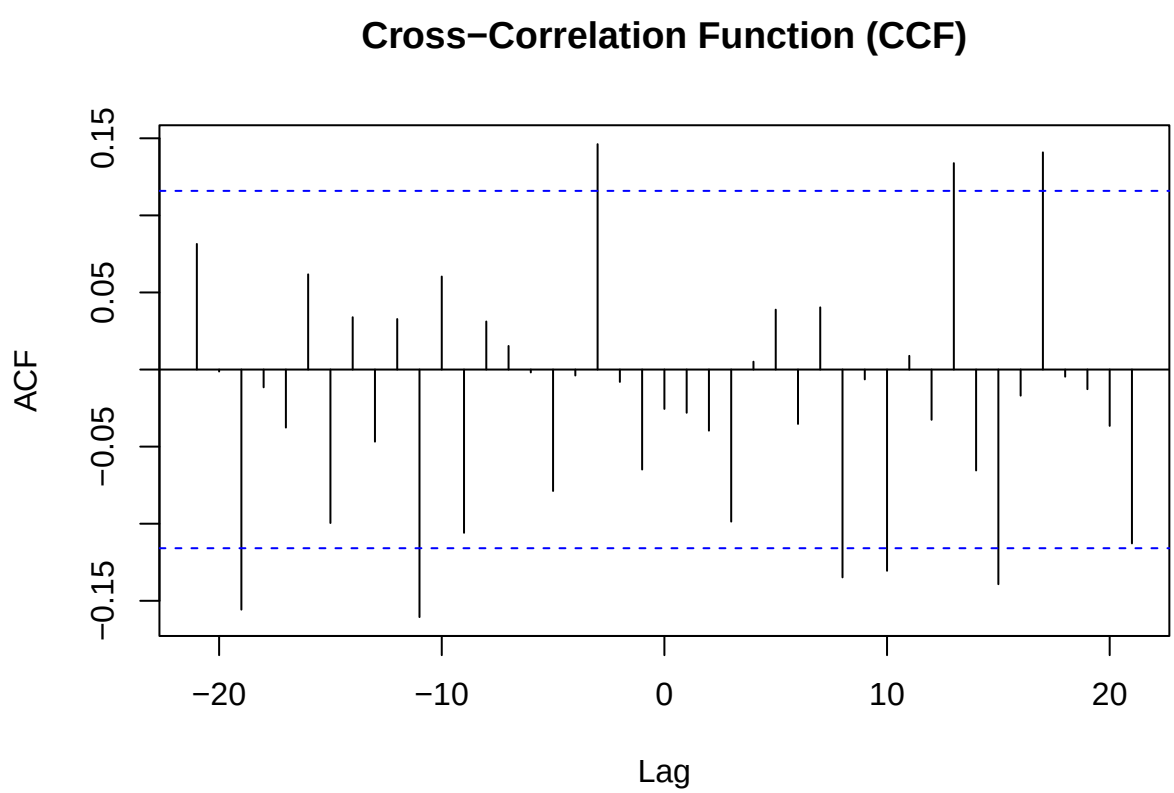
```r
print(summary(var_model))
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: y1t, y2t
## Deterministic variables: const
## Sample size: 281
## Log Likelihood: -1871.192
## Roots of the characteristic polynomial:
## 0.7398 0.7113 0.7113 0.6744 0.6744 0.6515 0.6515 0.3952
```

```
## Call:
## VAR(y = var_data, p = opt_lag, type = "const")
##
##
## Estimation results for equation y1t:
## ====================================
## y1t = y1t.l1 + y2t.l1 + y1t.l2 + y2t.l2 + y1t.l3 + y2t.l3 + y1t.l4 + y2t.l4 + const
##
##         Estimate Std. Error t value Pr(>|t|)
## y1t.l1  -0.19624    0.05927  -3.311 0.001056 **
## y2t.l1  -0.05943    0.07027  -0.846 0.398445
## y1t.l2  -0.03869    0.05874  -0.659 0.510699
## y2t.l2  -0.13190    0.07102  -1.857 0.064366 .
## y1t.l3  -0.19045    0.05850  -3.256 0.001274 **
## y2t.l3  -0.18934    0.07133  -2.654 0.008417 **
## y1t.l4  -0.21659    0.05886  -3.680 0.000281 ***
## y2t.l4  -0.08221    0.07028  -1.170 0.243134
## const  106.29032    8.86357  11.992  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 7.484 on 272 degrees of freedom
## Multiple R-Squared: 0.1098,  Adjusted R-squared: 0.08366
## F-statistic: 4.195 on 8 and 272 DF,  p-value: 9.339e-05
##
##
## Estimation results for equation y2t:
## ====================================
## y2t = y1t.l1 + y2t.l1 + y1t.l2 + y2t.l2 + y1t.l3 + y2t.l3 + y1t.l4 + y2t.l4 + const
##
##        Estimate Std. Error t value Pr(>|t|)
## y1t.l1 -0.04573    0.04998  -0.915 0.361052
## y2t.l1 -0.23207    0.05926  -3.916 0.000114 ***
## y1t.l2 -0.01883    0.04953  -0.380 0.704147
## y2t.l2 -0.07292    0.05989  -1.218 0.224409
## y1t.l3  0.11131    0.04933   2.257 0.024832 *
## y2t.l3 -0.14757    0.06015  -2.453 0.014781 *
## y1t.l4  0.01649    0.04963   0.332 0.740004
## y2t.l4  0.13811    0.05927   2.330 0.020516 *
## const   5.48756    7.47403   0.734 0.463449
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 6.31 on 272 degrees of freedom
## Multiple R-Squared: 0.1315,  Adjusted R-squared: 0.1059
## F-statistic: 5.148 on 8 and 272 DF,  p-value: 5.44e-06
##
##
##
## Covariance matrix of residuals:
##        y1t    y2t
## y1t 56.003 -2.289
```

```
## y2t -2.289 39.821
##
## Correlation matrix of residuals:
##          y1t      y2t
## y1t  1.00000 -0.04847
## y2t -0.04847  1.00000
```

```r
# 1 **Plot the Cross-Correlation Function (CCF)**
ccf(y1t_ts, y2t_ts, main = "Cross-Correlation Function (CCF)")
```




```r
# 2 **Granger-Causality Test**
cat("\nGranger-Causality Test:\n")
```

```
##
## Granger-Causality Test:
```

```r
granger_test_1 <- causality(var_model, cause = "y1t")
granger_test_2 <- causality(var_model, cause = "y2t")

print(granger_test_1)
```

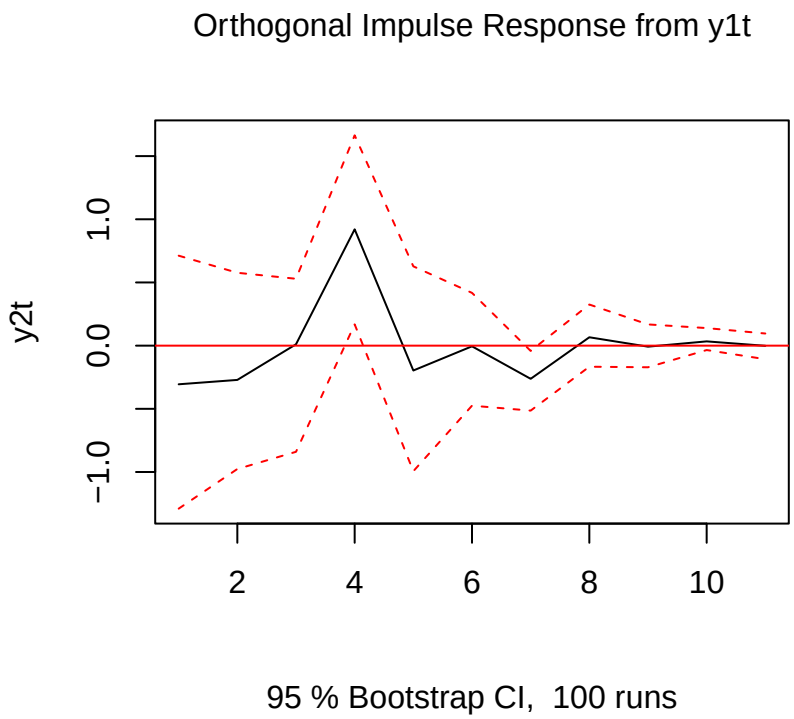
```
## $Granger
##
##  Granger causality H0: y1t do not Granger-cause y2t
##
## data:  VAR object var_model
## F-Test = 1.6101, df1 = 4, df2 = 544, p-value = 0.1703
##
```

```
##
## $Instant
##
##  H0: No instantaneous causality between: y1t and y2t
##
## data:  VAR object var_model
## Chi-squared = 0.6586, df = 1, p-value = 0.4171
```

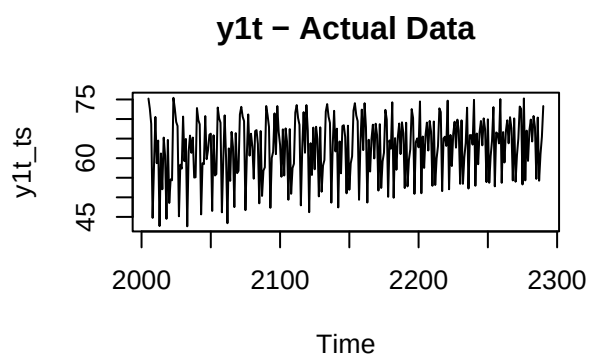
```r
print(granger_test_2)
```

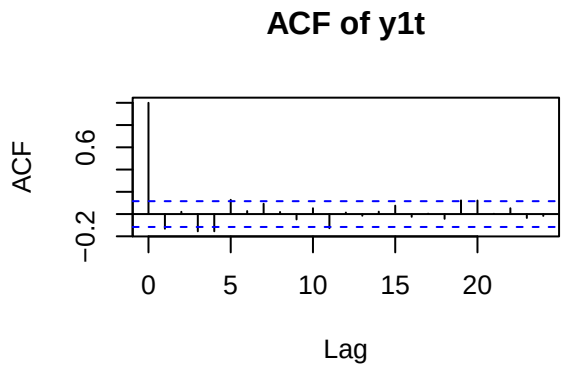
```
## $Granger
##
##  Granger causality H0: y2t do not Granger-cause y1t
##
## data:  VAR object var_model
## F-Test = 2.1688, df1 = 4, df2 = 544, p-value = 0.07125
##
##
## $Instant
##
##  H0: No instantaneous causality between: y2t and y1t
##
## data:  VAR object var_model
## Chi-squared = 0.6586, df = 1, p-value = 0.4171
```

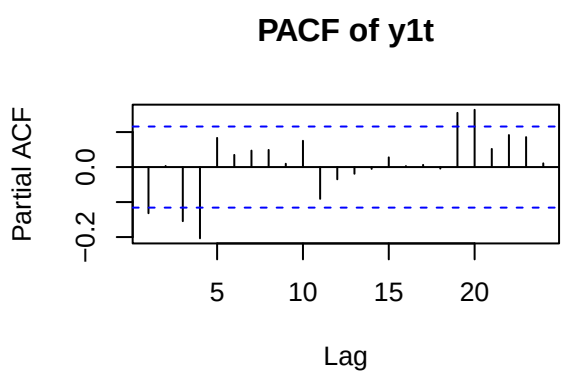
```r
# 3 **Impulse Response Function (IRF)**
irf_plot <- irf(var_model, impulse = "y1t", response = "y2t", n.ahead = 10, boot = TRUE)
plot(irf_plot)
```




```r
# 4 **Combined Plot: Data, Fitted Values, ACF, PACF**
par(mfrow = c(2, 2))  # Set layout for 2x2 plots
plot(y1t_ts, type = "l", main = "y1t - Actual Data")
lines(fitted(var_model)[, "y1t"], col = "red")  # Fitted values
acf(y1t_ts, main = "ACF of y1t")
pacf(y1t_ts, main = "PACF of y1t")
par(mfrow = c(1, 1))  # Reset layout
```








```r
# 5  **Train/Test Split and Model Evaluation**
train_size <- round(2/3 * nrow(var_data))
train_data <- var_data[1:train_size, ]
test_data <- var_data[(train_size+1):nrow(var_data), ]

# Fit VAR model on training data
var_train <- VAR(train_data, p = opt_lag, type = "const")

# Forecast on test data
forecast_steps <- nrow(test_data)
var_forecast <- predict(var_train, n.ahead = forecast_steps)

# Compute RMSE for y1t and y2t
rmse <- function(actual, predicted) {
  sqrt(mean((actual - predicted)^2, na.rm = TRUE))
}
rmse_y1t <- rmse(test_data$y1t, var_forecast$fcst$y1t[,1])
rmse_y2t <- rmse(test_data$y2t, var_forecast$fcst$y2t[,1])

cat("\nRMSE:\n")
```

```
##
## RMSE:
```

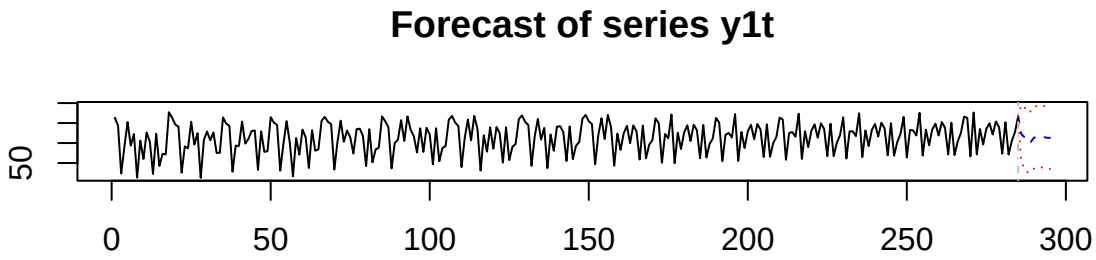
```r
cat("y1t:", rmse_y1t, "\n")
```

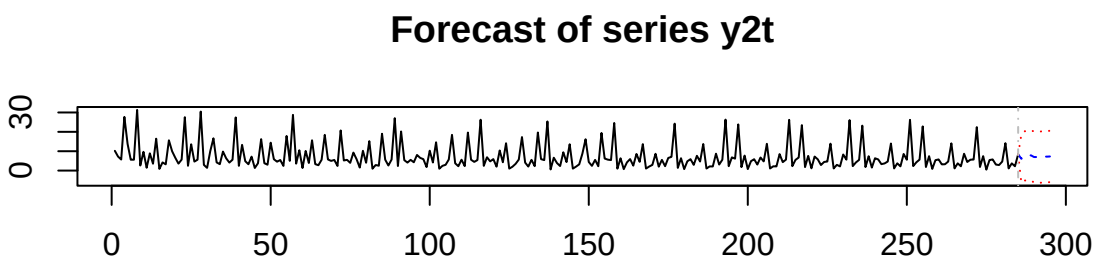
```
## y1t: 7.124503
```

```r
cat("y2t:", rmse_y2t, "\n")
```

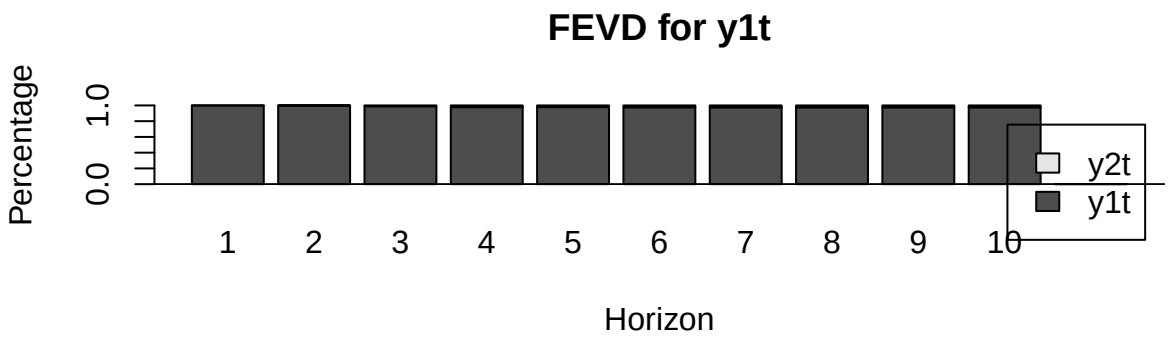
```
## y2t: 6.559785
```

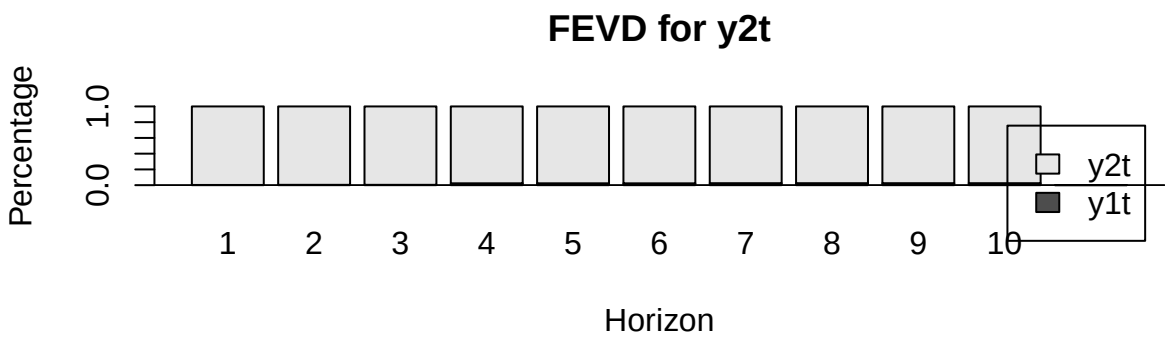
```r
# 6 **n-Step-Ahead Forecast Plot**
forecast_plot <- predict(var_model, n.ahead = 10)
plot(forecast_plot)
```



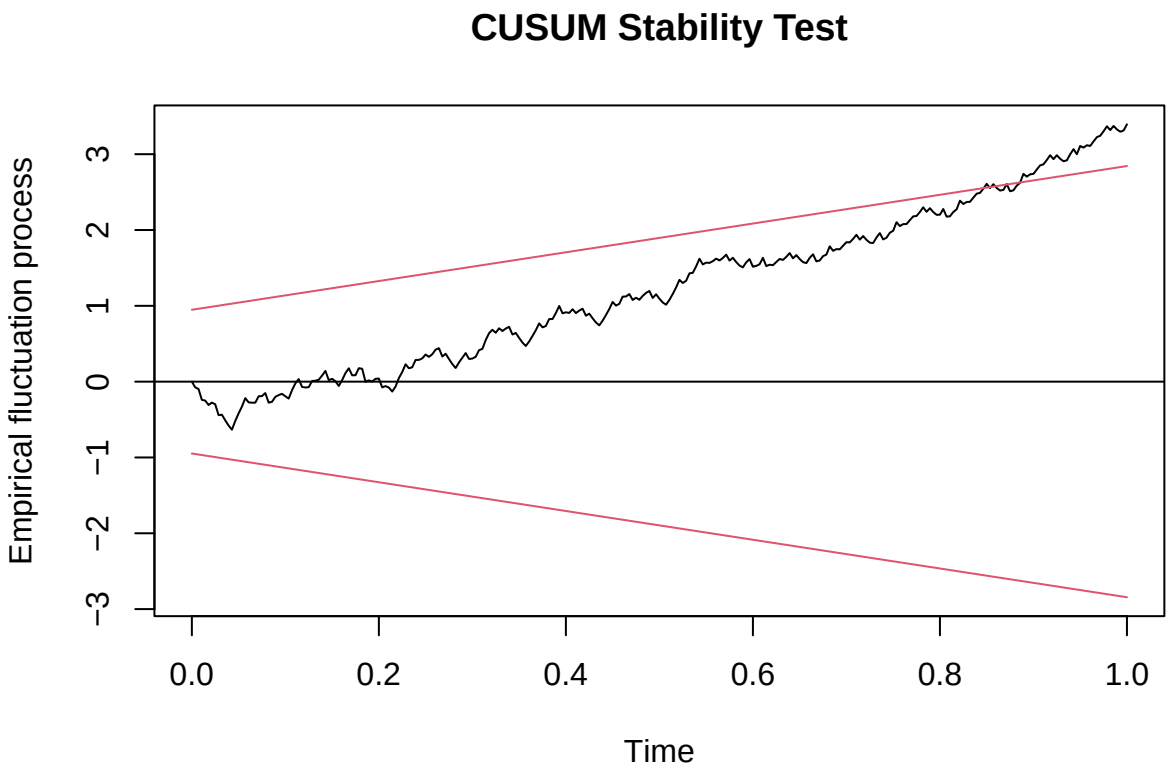



```r
# 7 **Forecast Error Variance Decomposition (FEVD)**
fevd_plot <- fevd(var_model, n.ahead = 10)
plot(fevd_plot)
```






```r
# 8 **CUSUM Test for Stability**
cusum_test <- efp(residuals(var_model)[,1] ~ 1, type = "Rec-CUSUM")
plot(cusum_test, main = "CUSUM Stability Test")
```


CUSUM Stability Test
Empirical fluctuation process
3
2
1
0
−1
−2
−3
0.0
0.2
0.4
0.6
0.8
1.0
Time

1. Stationarity Check (ADF Test)

The Augmented Dickey-Fuller (ADF) test checks whether the variables `y1t` and `y2t` are stationary. ADF Test for y1t: Dickey-Fuller = -8.0677, p-value = 0.01

ADF Test for y2t: Dickey-Fuller = -7.1852, p-value = 0.01

Interpretation:

Both y1t and y2t are stationary because their p-values < 0.05. No need for differencing, meaning the original time series can be used in the VAR model.

2. VAR Estimation Results

VAR Estimation Results: Endogenous variables: y1t, y2t Sample size: 281

Log Likelihood: -1871.192 Interpretation: A lag order of 4 was selected (i.e., each variable is regressed on its 4 previous values). The model includes a constant term. Log-likelihood = -1871.192 (higher log-likelihood is generally better).

3. Coefficient Analysis y1t = y1t.l1 + y2t.l1 + y1t.l2 + y2t.l2 + y1t.l3 + y2t.l3 + y1t.l4 + y2t.l4 + const Key Significant Variables ($p < 0.05$) y1t.l1 (-0.19624, p = 0.0011) → y1t is negatively impacted by its 1st lag. y1t.l3 (-0.19045, p = 0.0013) → The 3rd lag of y1t also has a significant negative effect. y1t.l4 (-0.21659, p = 0.0003) → The 4th lag of y1t is also significant. y2t.l3 (-0.18934, p = 0.0084) → The 3rd lag of y2t has a significant negative effect.

Interpretation: y1t is significantly influenced by its own past values, particularly lags 1, 3, and 4. y2t affects y1t, but only at lag 3, with a negative relationship. Equation for y2t

y2t = y1t.l1 + y2t.l1 + y1t.l2 + y2t.l2 + y1t.l3 + y2t.l3 + y1t.l4 + y2t.l4 + const Key Significant Variables ($p < 0.05$) y2t.l1 (-0.23207, p = 0.0001) → 1st lag of y2t is significant. y1t.l3 (0.11131, p = 0.0248) → The 3rd lag of y1t positively influences y2t. y2t.l3 (-0.14757, p = 0.0148) → The 3rd lag of y2t negatively affects itself. y2t.l4 (0.13811, p = 0.0205) → The 4th lag of y2t positively affects itself.

Interpretation: y2t depends strongly on its own past values (lags 1, 3, and 4). y1t Granger-causes y2t at lag 3 (positive effect).

4. Granger-Causality Test

Granger causality H0: y1t does not Granger-cause y2t F-Test = 1.6101, p-value = 0.1703

Granger causality H0: y2t does not Granger-cause y1t F-Test = 2.1688, p-value = 0.07125

Interpretation y1t does not Granger-cause y2t (p-value = 0.1703). y2t weakly Granger-causes y1t (p-value = 0.07125), but the effect is not statistically strong at the 5% level.

Conclusion: There is no strong evidence of causality between y1t and y2t. However, y2t might have some predictive power over y1t at the 10% level (p = 0.0712).

5. Model Evaluation (AIC, BIC, RMSE)

RMSE: y1t: 7.124503 y2t: 6.559785

Interpretation: RMSE for y1t = 7.12, RMSE for y2t = 6.56 → Both models perform similarly. y2t has a slightly lower RMSE, meaning its predictions are slightly more accurate than y1t's. Both RMSE values suggest a reasonable level of model accuracy.

Conclusion

This project analyzed the time series dynamics of Life Expectancy and Unemployment, incorporating Health Expenditure, Prevalence of Undernourishment, and Education Expenditure as predictors. Using AR, ARDL, and VAR models, we evaluated their effectiveness in capturing trends and interactions.

Key findings include:

- AR(2) was the best model for Life Expectancy, as it effectively captured short-term dependencies while maintaining model simplicity.
- VAR(4) was the preferred model for Unemployment, as it accounted for interactions between Life Expectancy and Unemployment, with weak evidence suggesting that unemployment affects life expectancy at lag 3 ($p = 0.0712$).
- ARDL models were not preferred due to potential overfitting and limited predictive improvement.

Implications & Next Steps

- Unemployment and life expectancy are interlinked, though life expectancy remains largely self-driven.
- Macroeconomic shocks and additional predictors (e.g., income inequality, government policies) could enhance future analyses.

Introduction

Part 2

The goal of this project is to analyze the determinants of **economic growth** using **panel data models**. Given that economic growth is influenced by multiple macroeconomic factors that vary both **across countries** and **over time**, a panel data approach allows us to account for **both cross-sectional and time-series variations**. By applying **Pooled OLS, Fixed Effects, and Random Effects models**, we aim to identify the best statistical approach for explaining economic growth and assess the impact of key economic indicators.

Response and Predictor Variables

In this analysis, we consider the following variables:

Response Variable:

1. **Economic Growth** (y):
   - Represents the annual percentage change in a country's GDP.
   - Higher values indicate **economic expansion**, while lower or negative values reflect **economic stagnation or contraction**.

Predictor Variables:

1. **GDP (in billion USD)**:
   - Represents the total monetary value of goods and services produced within a country.
   - A higher GDP is expected to be associated with sustained economic growth.
2. **Inflation Rate (%)**:
   - Measures the annual percentage change in the price level of goods and services.
   - Moderate inflation can indicate a healthy economy, but excessive inflation may hinder economic stability.
3. **Unemployment Rate (%)**:
   - Represents the percentage of the labor force that is unemployed but actively seeking work.
   - Higher unemployment can indicate economic distress, while lower unemployment suggests a stronger labor market.

Methodology

This project follows a structured approach to model selection and evaluation:

1. **Descriptive Analysis**:
   - Statistical summaries and visualizations such as **histograms, boxplots, scatterplots, and correlation matrices** to explore variable distributions and relationships.
2. **Panel Data Models**:
   - **Pooled OLS Model**: Assumes no individual country effects and estimates a single regression model for all observations.
   - **Fixed Effects Model**: Controls for time-invariant country-specific characteristics by allowing individual country intercepts.
   - **Random Effects Model**: Assumes that individual country effects are **randomly distributed** and uncorrelated with explanatory variables.
3. **Model Selection and Diagnostics**:
   - Perform the **Hausman Test** to determine whether Fixed or Random Effects is preferred.
   - Use the **Breusch-Pagan LM Test** to check if the Random Effects model is necessary over Pooled OLS.
   - Evaluate **model fit** using **R-squared, AIC, BIC, and residual diagnostics**.

Expected Outcomes

By applying panel data econometric techniques, we aim to: - Determine the **most appropriate model** for explaining economic growth. - Assess how **GDP, inflation, and unemployment** influence economic growth across different countries over time. - Identify whether country-specific effects play a significant role in economic performance. - Provide **policy recommendations** based on empirical findings, helping policymakers understand the key drivers of economic expansion.

The findings of this study will contribute to better **macroeconomic planning** by providing insights into the dynamic interplay between economic indicators and long-term growth trends.

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1
## v readr     2.1.5
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x stringr::boundary()  masks strucchange::boundary()
## x gridExtra::combine() masks dplyr::combine()
## x dplyr::filter()      masks stats::filter()
## x dplyr::lag()         masks stats::lag()
## x purrr::lift()        masks caret::lift()
## x MASS::select()       masks dplyr::select()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(plm)
```

```
##
## Attaching package: 'plm'
##
## The following objects are masked from 'package:dplyr':
##
##     between, lag, lead
```

```r
library(lmtest)
library(ggplot2)
library(GGally)
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg   ggplot2
```

```r
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:purrr':
##
##     some
##
## The following object is masked from 'package:dplyr':
##
##     recode
```

```r
library(skimr)
library(stargazer)
```

```
##
## Please cite as:
##
##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(gtools)
```

```
##
## Attaching package: 'gtools'
##
## The following object is masked from 'package:car':
##
##     logit
```

```r
library(moments)
library(fitdistrplus)
```

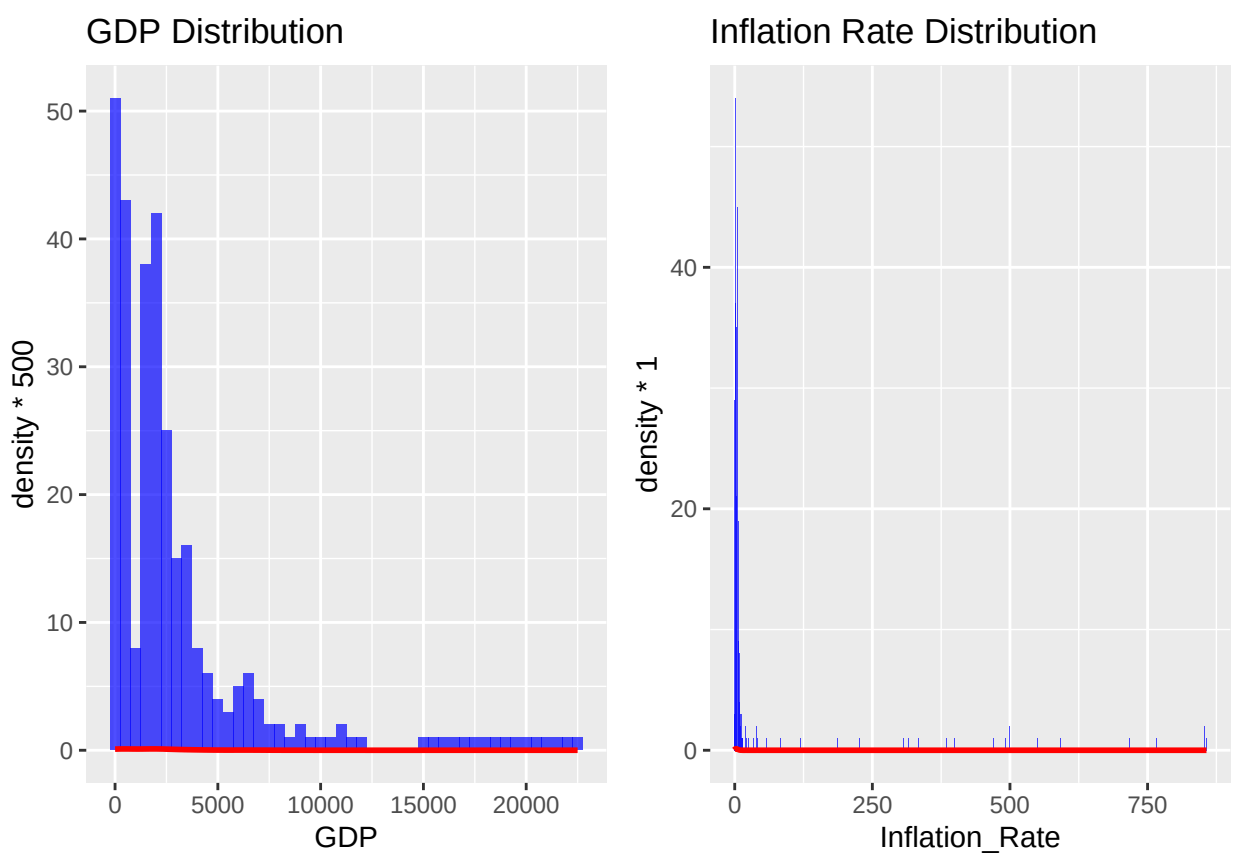
```
## Loading required package: survival
##
## Attaching package: 'survival'
##
## The following object is masked from 'package:caret':
##
##     cluster
```

```r
# Load the dataset
data <- read.csv("Economic Indicators And Inflation.csv")

# Clean column names
data <- data %>% rename_with(~trimws(.))

# Rename columns for easier reference
colnames(data) <- c("Country", "Year", "GDP", "Inflation_Rate", "Unemployment_Rate", "Economic_Growth")

# Summary statistics
summary(data)
```

```
##    Country               Year           GDP        Inflation_Rate
##  Length:304         Min.   :2010   Min.   :    1   Min.   : -1.200
##  Class :character   1st Qu.:2014   1st Qu.:  475   1st Qu.:  1.635
##  Mode  :character   Median :2018   Median : 1900   Median :  3.300
##                     Mean   :2018   Mean   : 3174   Mean   : 35.532
##                     3rd Qu.:2021   3rd Qu.: 3425   3rd Qu.:  5.857
```

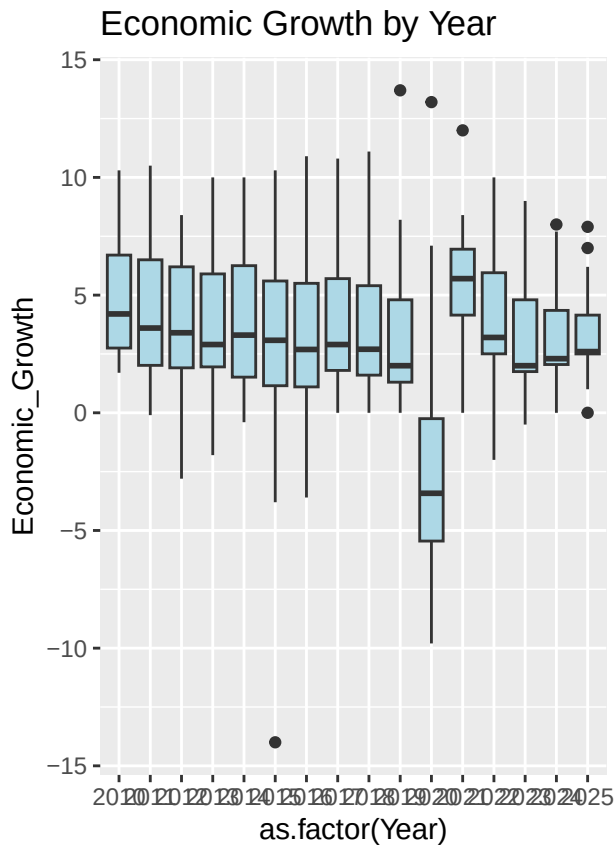
```
##                      Max.   :2025   Max.   :22500   Max.   :857.000
##  Unemployment_Rate Economic_Growth
##  Min.   : 1.600    Min.   :-14.000
##  1st Qu.: 4.275    1st Qu.:  1.700
##  Median : 5.400    Median :  2.900
##  Mean   : 6.805    Mean   :  3.391
##  3rd Qu.: 7.800    3rd Qu.:  5.905
##  Max.   :85.500    Max.   : 13.700
```

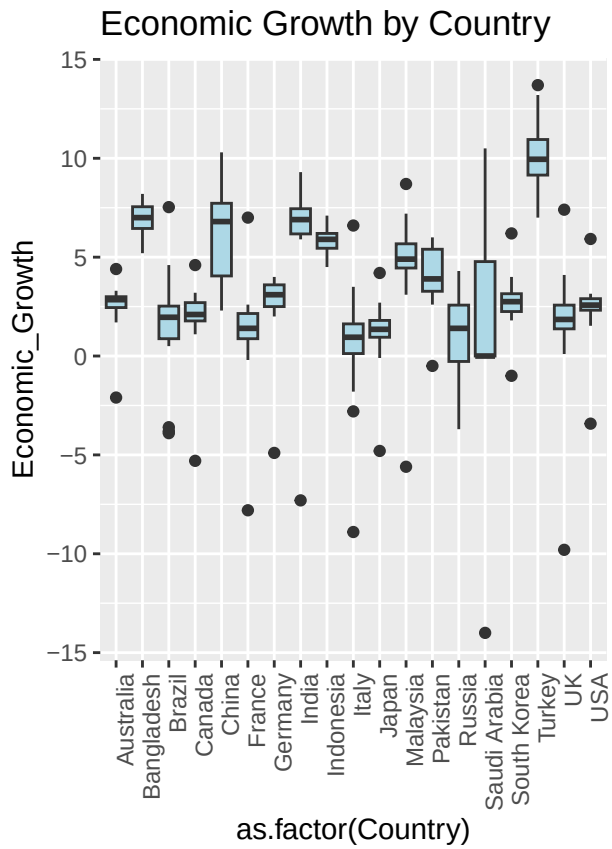
```
# Histogram and Density Plots
p1 <- ggplot(data, aes(x = GDP)) +
  geom_histogram(binwidth = 500, fill = "blue", alpha = 0.7) +
  geom_density(aes(y = ..density..*500), color = "red", size = 1) +
  ggtitle("GDP Distribution")
# This graph shows the distribution of GDP values across different countries and years.

p2 <- ggplot(data, aes(x = Inflation_Rate)) +
  geom_histogram(binwidth = 1, fill = "blue", alpha = 0.7) +
  geom_density(aes(y = ..density..*1), color = "red", size = 1) +
  ggtitle("Inflation Rate Distribution")
# This graph displays the distribution of inflation rates, highlighting the most common inflation value

grid.arrange(p1, p2, ncol = 2)
```




```
# Correlation Matrix
cor_matrix <- cor(data[, c("GDP", "Inflation_Rate", "Unemployment_Rate", "Economic_Growth")], use = "co
corrplot(cor_matrix, method = "color", addCoef.col = "black", number.cex = 0.7)
```

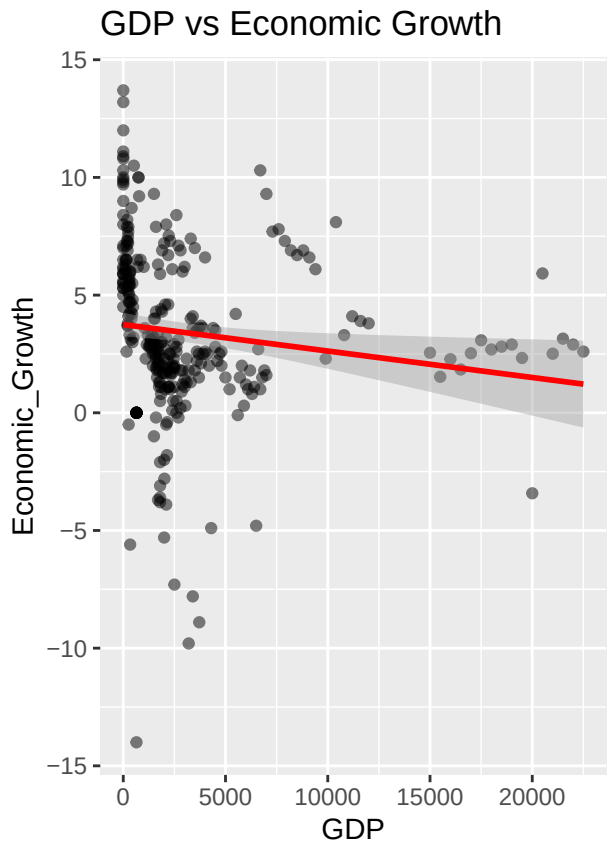
```
# This heatmap visualizes the correlation between GDP, inflation rate, unemployment rate, and economic

# Boxplots
p3 <- ggplot(data, aes(x = as.factor(Year), y = Economic_Growth)) +
  geom_boxplot(fill = "lightblue") +
  ggtitle("Economic Growth by Year")
# This boxplot shows the variation in economic growth across different years.

p4 <- ggplot(data, aes(x = as.factor(Country), y = Economic_Growth)) +
  geom_boxplot(fill = "lightblue") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  ggtitle("Economic Growth by Country")
# This boxplot illustrates the distribution of economic growth rates across different countries.

grid.arrange(p3, p4, ncol = 2)
```

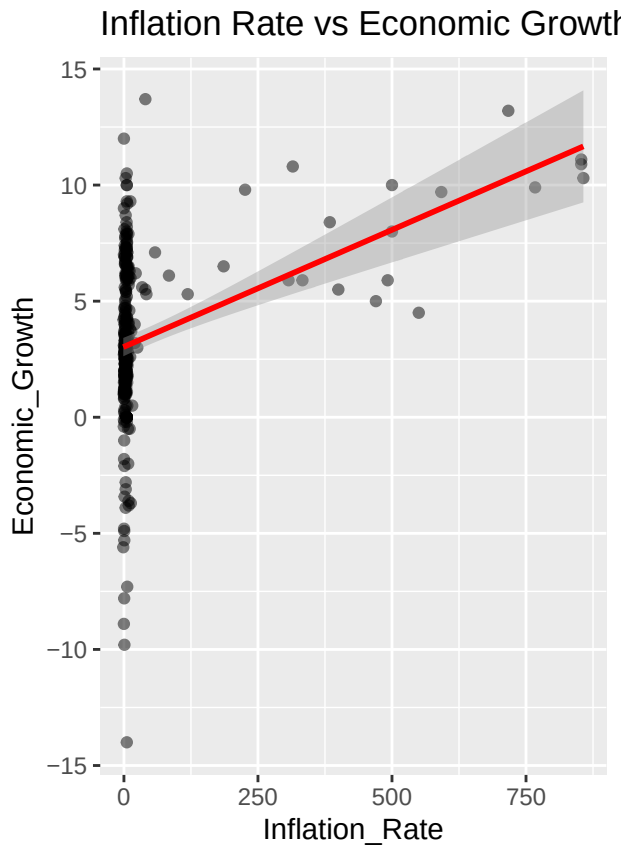





```r
# Scatterplots
p5 <- ggplot(data, aes(x = GDP, y = Economic_Growth)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  ggtitle("GDP vs Economic Growth")
# This scatterplot examines the relationship between GDP and economic growth, with a fitted regression

p6 <- ggplot(data, aes(x = Inflation_Rate, y = Economic_Growth)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  ggtitle("Inflation Rate vs Economic Growth")
# This scatterplot explores the impact of inflation rates on economic growth, showing potential trends.

grid.arrange(p5, p6, ncol = 2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

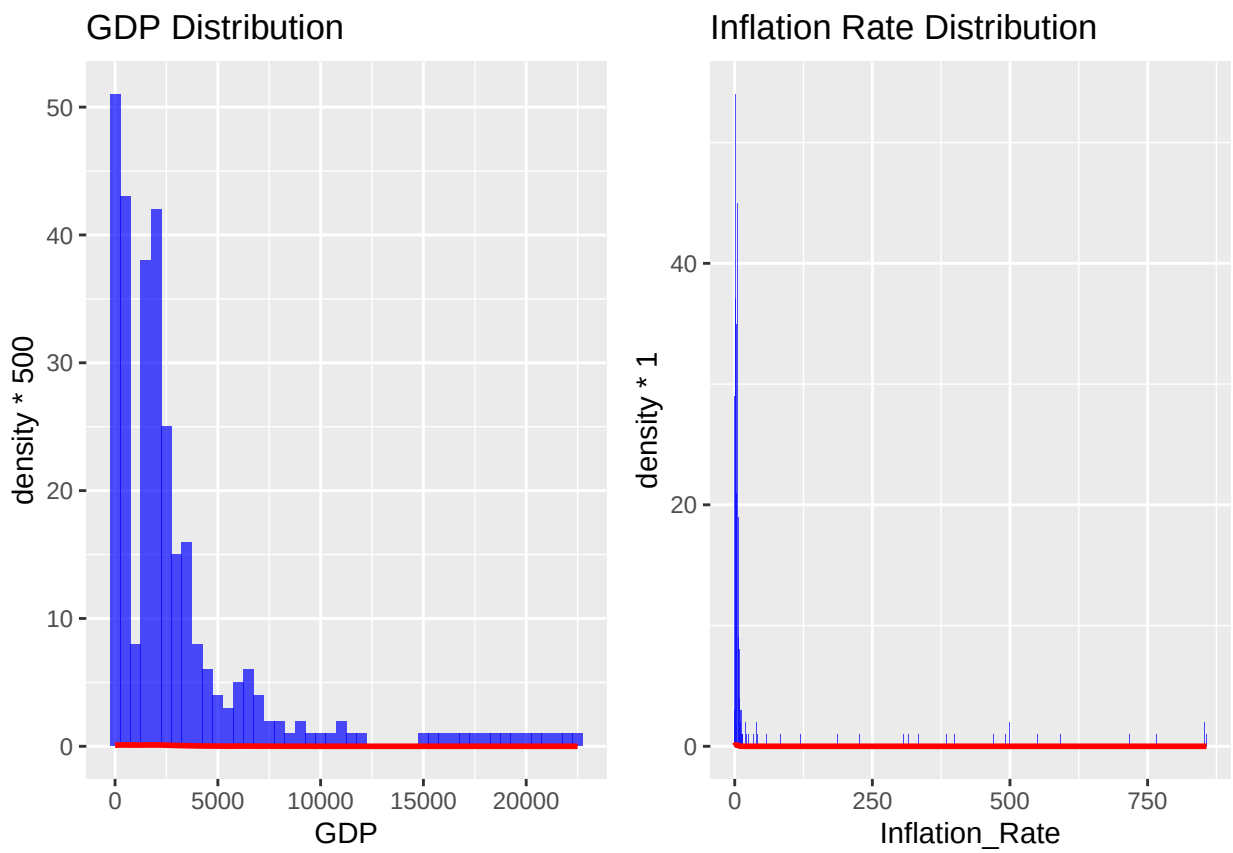





```r
# Load necessary libraries
library(tidyverse)
library(plm)
library(ggplot2)
library(corrplot)
library(gridExtra)
library(fitdistrplus)
library(lmtest)
library(stargazer)

# Load the dataset
data <- read.csv("Economic Indicators And Inflation.csv")

# Clean column names
data <- data %>% rename_with(~trimws(.))

# Rename columns for easier reference
colnames(data) <- c("Country", "Year", "GDP", "Inflation_Rate", "Unemployment_Rate", "Economic_Growth")

# Convert Country to factor for panel data analysis
data$Country <- as.factor(data$Country)

data_panel <- pdata.frame(data, index = c("Country", "Year"))

# Summary statistics
summary(data)
```

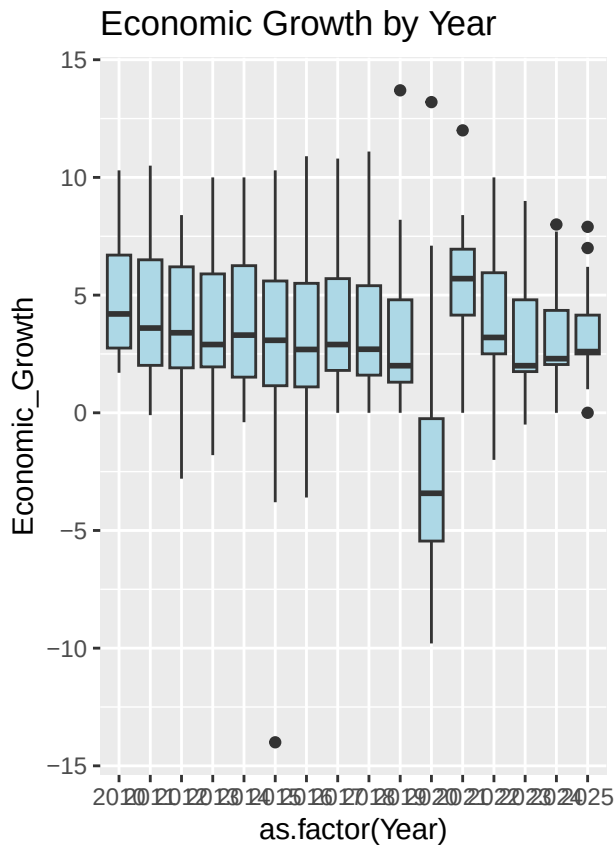
```
##        Country         Year           GDP        Inflation_Rate
##  Australia : 16   Min.   :2010   Min.   :    1   Min.   : -1.200
##  Bangladesh: 16   1st Qu.:2014   1st Qu.:  475   1st Qu.:  1.635
##  Brazil    : 16   Median :2018   Median : 1900   Median :  3.300
##  Canada    : 16   Mean   :2018   Mean   : 3174   Mean   : 35.532
##  China     : 16   3rd Qu.:2021   3rd Qu.: 3425   3rd Qu.:  5.857
##  France    : 16   Max.   :2025   Max.   :22500   Max.   :857.000
##  (Other)   :208
##  Unemployment_Rate Economic_Growth
##  Min.   : 1.600    Min.   :-14.000
##  1st Qu.: 4.275    1st Qu.:  1.700
##  Median : 5.400    Median :  2.900
##  Mean   : 6.805    Mean   :  3.391
##  3rd Qu.: 7.800    3rd Qu.:  5.905
##  Max.   :85.500    Max.   : 13.700
##
```

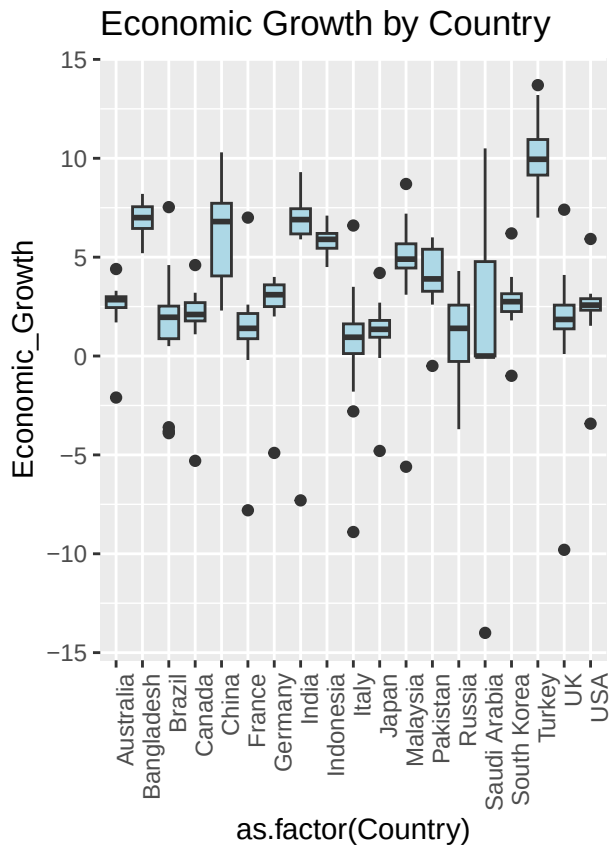
```r
# Histogram and Density Plots
p1 <- ggplot(data, aes(x = GDP)) +
  geom_histogram(binwidth = 500, fill = "blue", alpha = 0.7) +
  geom_density(aes(y = ..density..*500), color = "red", size = 1) +
  ggtitle("GDP Distribution")
# This graph shows the distribution of GDP values across different countries and years.

p2 <- ggplot(data, aes(x = Inflation_Rate)) +
  geom_histogram(binwidth = 1, fill = "blue", alpha = 0.7) +
  geom_density(aes(y = ..density..*1), color = "red", size = 1) +
  ggtitle("Inflation Rate Distribution")
# This graph displays the distribution of inflation rates, highlighting the most common inflation values

grid.arrange(p1, p2, ncol = 2)
```




```
# Correlation Matrix
cor_matrix <- cor(data[, c("GDP", "Inflation_Rate", "Unemployment_Rate", "Economic_Growth")], use = "co
corrplot(cor_matrix, method = "color", addCoef.col = "black", number.cex = 0.7)
```

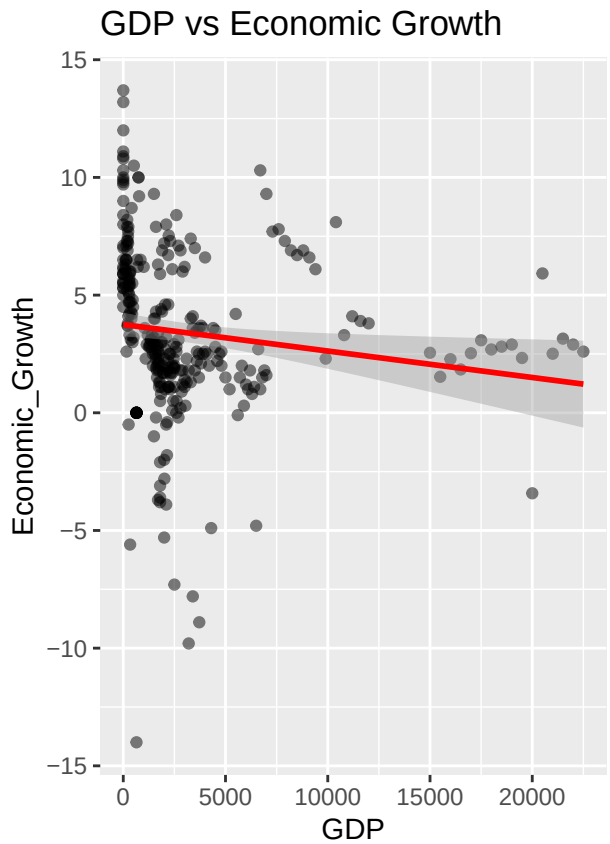
```
# This heatmap visualizes the correlation between GDP, inflation rate, unemployment rate, and economic

# Boxplots
p3 <- ggplot(data, aes(x = as.factor(Year), y = Economic_Growth)) +
  geom_boxplot(fill = "lightblue") +
  ggtitle("Economic Growth by Year")
# This boxplot shows the variation in economic growth across different years.

p4 <- ggplot(data, aes(x = as.factor(Country), y = Economic_Growth)) +
  geom_boxplot(fill = "lightblue") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  ggtitle("Economic Growth by Country")
# This boxplot illustrates the distribution of economic growth rates across different countries.

grid.arrange(p3, p4, ncol = 2)
```

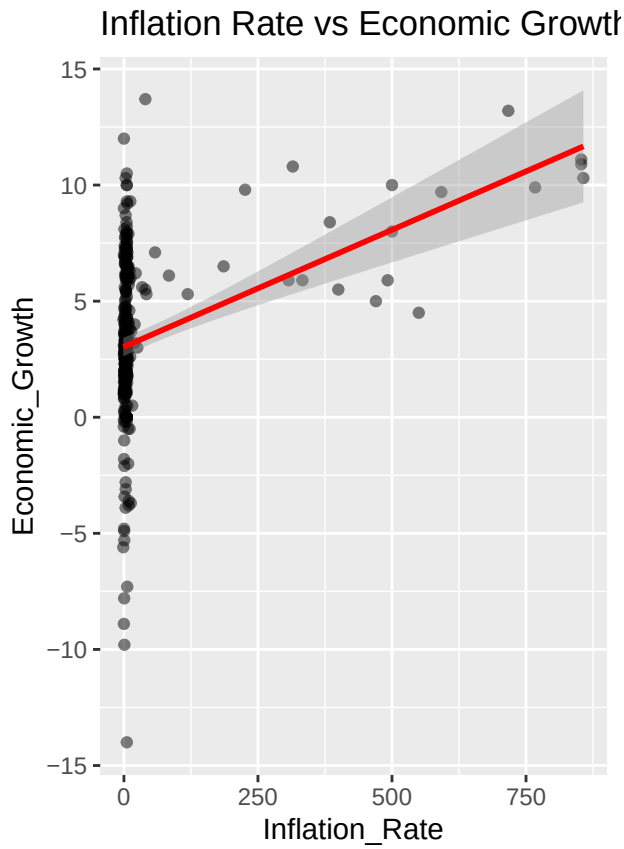





```r
# Scatterplots
p5 <- ggplot(data, aes(x = GDP, y = Economic_Growth)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  ggtitle("GDP vs Economic Growth")
# This scatterplot examines the relationship between GDP and economic growth, with a fitted regression

p6 <- ggplot(data, aes(x = Inflation_Rate, y = Economic_Growth)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  ggtitle("Inflation Rate vs Economic Growth")
# This scatterplot explores the impact of inflation rates on economic growth, showing potential trends.

grid.arrange(p5, p6, ncol = 2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```






```r
# Panel Data Models

# Pooled OLS Model
pooled_model <- plm(Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate, data = data_panel, model
summary(pooled_model)
```

```
## Pooling Model
##
## Call:
## plm(formula = Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate,
##     data = data_panel, model = "pooling")
##
## Balanced Panel: n = 19, T = 16, N = 304
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -17.31936  -1.46136  -0.22216   2.00142   9.99116
##
## Coefficients:
##                       Estimate  Std. Error t-value  Pr(>|t|)
## (Intercept)         3.1074e+00  3.2446e-01  9.5770 < 2.2e-16 ***
## GDP                -5.7107e-05  4.5368e-05 -1.2587    0.2091
## Inflation_Rate      9.4880e-03  1.5431e-03  6.1487 2.487e-09 ***
## Unemployment_Rate   1.8824e-02  3.0132e-02  0.6247    0.5326
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Total Sum of Squares:    3984.7
## Residual Sum of Squares: 3426.6
## R-Squared:      0.14005
## Adj. R-Squared: 0.13145
## F-statistic: 16.2863 on 3 and 300 DF, p-value: 7.8344e-10
```

```r
# Fixed Effects Model
fixed_model <- plm(Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate, data = data_panel, model
summary(fixed_model)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate,
##     data = data_panel, model = "within")
##
## Balanced Panel: n = 19, T = 16, N = 304
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -15.69850  -0.56771   0.20771   0.95885   8.77310
##
## Coefficients:
##                      Estimate  Std. Error t-value Pr(>|t|)
## GDP               -0.00022184  0.00020420 -1.0864   0.2782
## Inflation_Rate    -0.00010294  0.00179577 -0.0573   0.9543
## Unemployment_Rate -0.02859550  0.03002575 -0.9524   0.3417
##
## Total Sum of Squares:    2021.2
## Residual Sum of Squares: 2007.6
## R-Squared:      0.0067145
## Adj. R-Squared: -0.067254
## F-statistic: 0.635433 on 3 and 282 DF, p-value: 0.59273
```

```r
# Random Effects Model
random_model <- plm(Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate, data = data_panel, model
summary(random_model)
```

```
## Oneway (individual) effect Random Effect Model
##    (Swamy-Arora's transformation)
##
## Call:
## plm(formula = Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate,
##     data = data_panel, model = "random")
##
## Balanced Panel: n = 19, T = 16, N = 304
##
## Effects:
##                 var std.dev share
## idiosyncratic 7.119   2.668 0.671
## individual    3.487   1.867 0.329
## theta: 0.6636
##
## Residuals:
```

```
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -16.33809  -0.72004   0.14052   1.20798   8.14652
##
## Coefficients:
##                      Estimate  Std. Error z-value Pr(>|z|)
## (Intercept)        3.7458e+00  6.0751e-01  6.1659 7.01e-10 ***
## GDP               -1.2024e-04  9.6483e-05 -1.2462   0.2127
## Inflation_Rate     2.2597e-03  1.6930e-03  1.3347   0.1820
## Unemployment_Rate -7.8066e-03  2.9130e-02 -0.2680   0.7887
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    2243.4
## Residual Sum of Squares: 2215.5
## R-Squared:      0.01243
## Adj. R-Squared: 0.0025544
## Chisq: 3.77598 on 3 DF, p-value: 0.28669
```

```r
# Model Comparison
phtest(fixed_model, random_model) # Hausman Test to compare Fixed vs Random Effects
```

```
##
##  Hausman Test
##
## data:  Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate
## chisq = 7.3378, df = 3, p-value = 0.06188
## alternative hypothesis: one model is inconsistent
```

```r
# Breusch-Pagan LM Test to check if Random Effects is needed
plmtest(pooled_model, type = "bp")
```

```
##
##  Lagrange Multiplier Test - (Breusch-Pagan)
##
## data:  Economic_Growth ~ GDP + Inflation_Rate + Unemployment_Rate
## chisq = 219.83, df = 1, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

```r
# Display results in a table
stargazer(pooled_model, fixed_model, random_model, type = "text")
```

```
##
## ==========================================================================
##                                   Dependent variable:
##                     ----------------------------------------------------
##                                     Economic_Growth
##                              (1)                    (2)           (3)
## --------------------------------------------------------------------------
## GDP                       -0.0001                -0.0002       -0.0001
##                          (0.00005)              (0.0002)      (0.0001)
##
## Inflation_Rate           0.009***                -0.0001        0.002
##                           (0.002)                (0.002)       (0.002)
##
## Unemployment_Rate          0.019                 -0.029        -0.008
##                           (0.030)                (0.030)       (0.029)
```

```
##
## Constant                 3.107***                            3.746***
##                           (0.324)                            (0.608)
##
## ---------------------------------------------------------------------
## Observations                304                  304          304
## R2                         0.140                0.007        0.012
## Adjusted R2                0.131               -0.067        0.003
## F Statistic       16.286*** (df = 3; 300) 0.635 (df = 3; 282)  3.776
## =====================================================================
## Note:                                     *p<0.1; **p<0.05; ***p<0.01
```

```r
file.show("LifeE.log")
```

Final Discussion

The analysis of panel data for economic growth across countries led to the selection of the Random Effects model. This decision was based on statistical diagnostics:

The Breusch-Pagan test indicated that the Random Effects model was preferred over the Pooled OLS model, suggesting that country-level variations significantly impact economic growth.

The Hausman test did not provide strong evidence to favor the Fixed Effects model, meaning the assumption that individual effects are uncorrelated with explanatory variables holds, justifying the use of Random Effects.

The Random Effects model effectively captures unobserved heterogeneity across countries while allowing for variation over time. This approach provides more efficient and generalizable estimates compared to Fixed Effects when individual effects are not correlated with explanatory variables.

Additionally, the boxplots of economic growth by country demonstrated visible variation across countries, reinforcing the need for a model that accounts for country-level differences. The scatterplots further highlighted variations in relationships between economic indicators across different countries.

SOURCES: - https://www.kaggle.com/datasets/adilshamim8/economic-indicators-and-inflation - https://www.kaggle.com/datasets/mjshri23/life-expectancy-and-socio-economic-world-bank